



Process Safety Engineering

Second unit: Hazard Analysis & Risk Assessment

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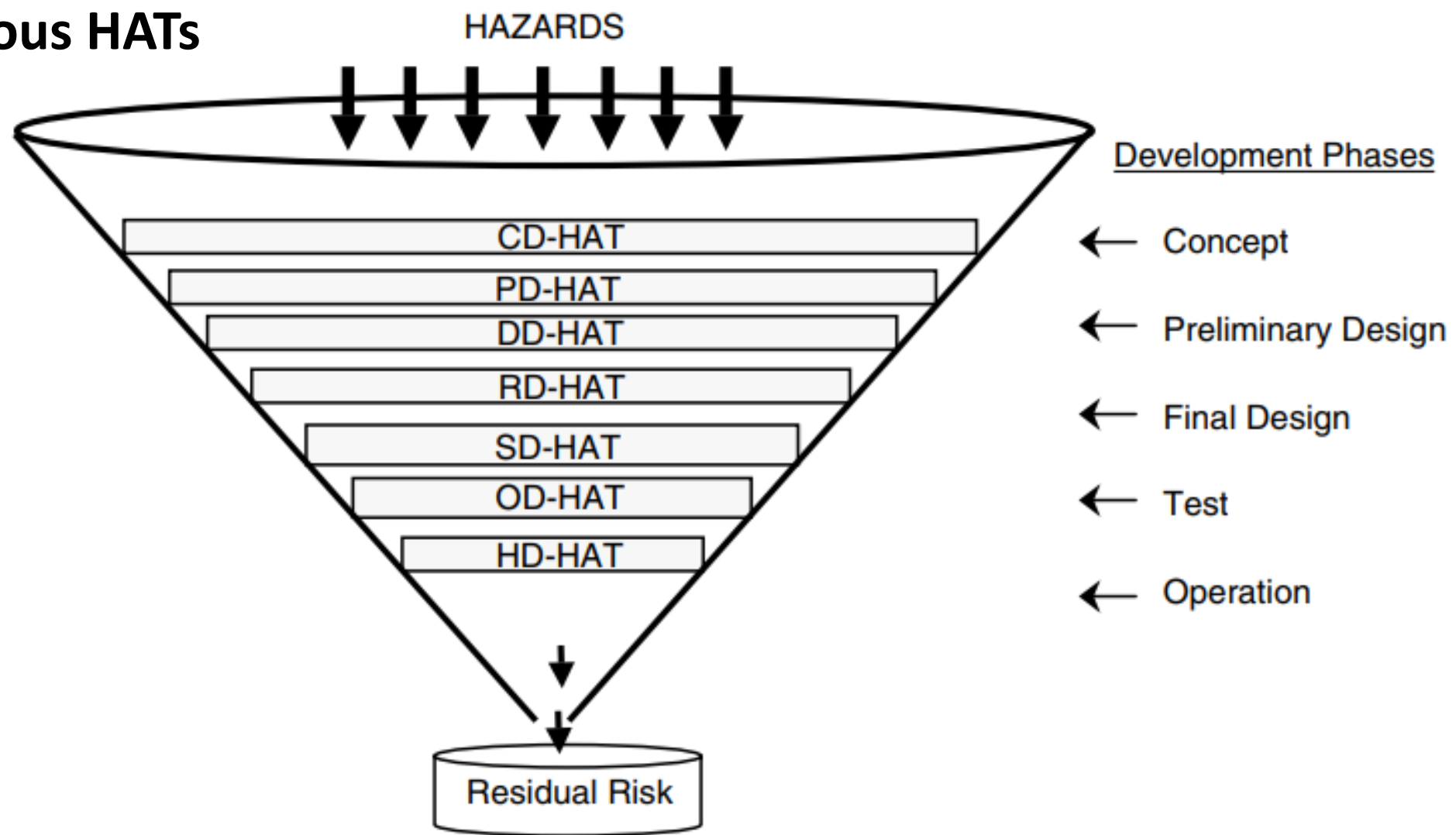
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Let us understand Hazard Analysis

- **Why?** - to identify hazards, hazard effects, & hazard causal factors
- **Ultimate goal?**
- **Performed to systematically examine the system, subsystem, facility, components, software, personnel, & their interrelationships**
- **2 categories of hazard analyses: types & techniques**
- **Type defines an analysis **category** (e.g., detailed design analysis), & technique defines a unique analysis **methodology** (e.g., fault tree analysis)**

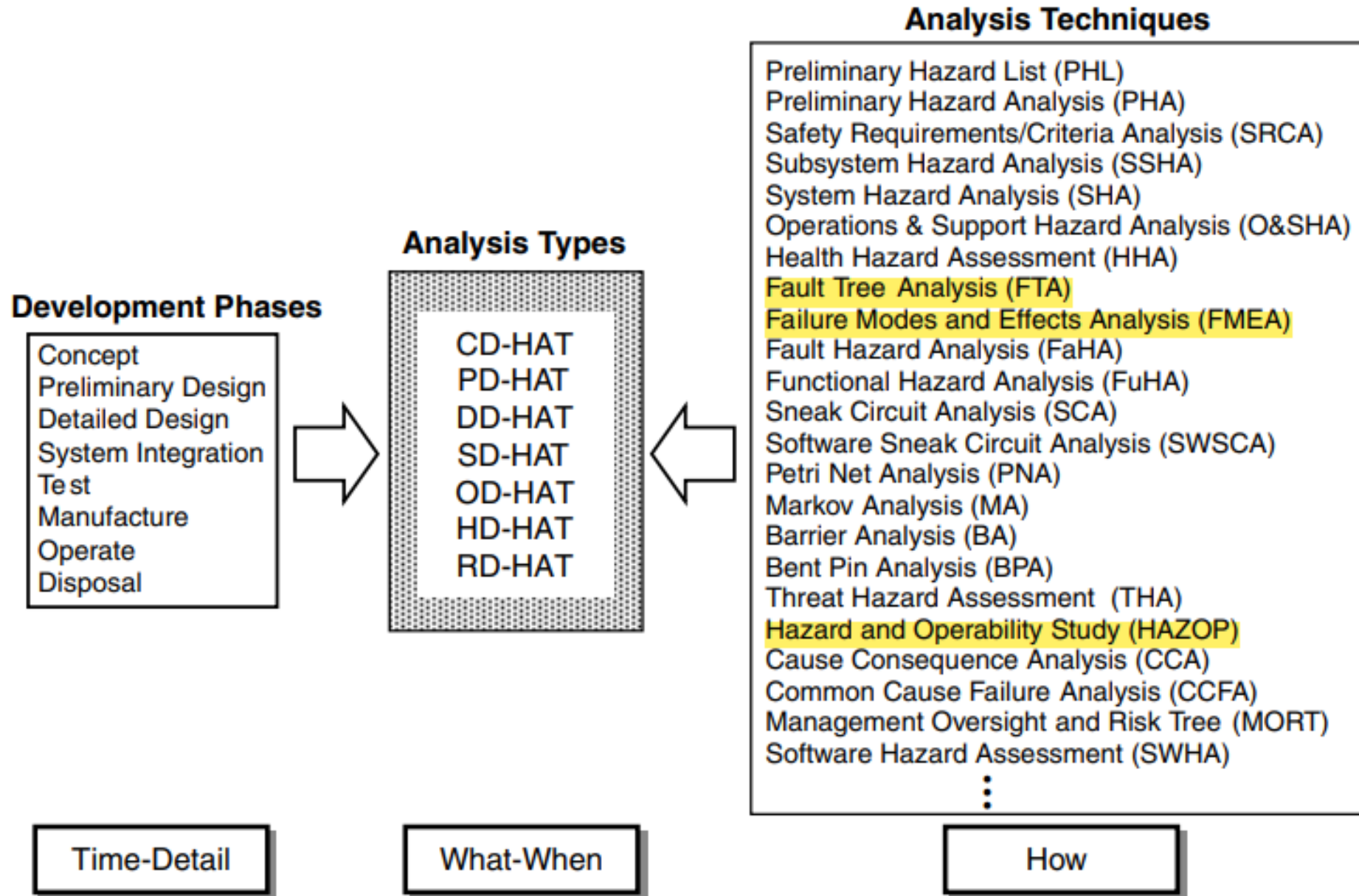
Preferred & referred book for Hazard Analysis: Ericson, C. A. (2015). *Hazard analysis techniques for system safety*. John Wiley & Sons

Hazard filters – various HATs



- 1 particular HAT does not necessarily identify all the hazards within a system
- Each HAT acts like a filter that identifies certain types of hazards

System life cycle & various hazard analyses



Major Attributes of Analysis Techniques

- ✓ Qualitative/quantitative
- ✓ Level of detail: design detail that can be evaluated by the technique
- ✓ Data required: type & level of design data required for the technique
- ✓ Program timing: effective time during system development for the technique
- ✓ Time required: relative amount of time required for the analysis
- ✓ Inductive/deductive
- ✓ Complexity
- ✓ Difficulty
- ✓ Technical expertise
- ✓ Tools required: technique is standalone or additional tools are necessary
- ✓ Cost
- ✓ Primary safety tool: a primary or secondary safety tool

Attributes of Analysis Techniques

Technique	Type	Identify Hazards	Identify Root Causes	Life-Cycle Phase	Qualitative/ Quantitative	Skill	Level of Detail	I/D
PHL	CD-HAT	Y	N	CD-PD	Qual.	SS	Minimal	I
PHA	PD-HAT	Y	P	CD-PD	Qual.	SS	Moderate to in-depth	I-D
SSHA	DD-HAT	Y	Y	DD	Qual.	SS, Engr., M&S	In-depth	I-D
SHA	SD-HAT	Y	Y	PD-DD-T	Qual.	SS, Engr., M&S	In-depth	I-D
O&SHA	OD-HAT	Y	Y	PD-DD-T	Qual.	SS, Engr., M&S	In-depth	I-D
HHA	HD-HAT	Y	Y	PD-DD-T	Qual.	SS, Engr., M&S	In-depth	I-D
SRCA	RD-HAT	P	N	PD-DD	Qual.	SS	In-depth	N/A
FTA	SD-HAT, DD-HAT	P	Y	PD-DD	Qual./Quant.	SS, Engr., M&S	Moderate to in-depth	D
ETA	SD-HAT	P	P	PD-DD	Qual./Quant.	SS, Engr., M&S	Moderate to in-depth	D
FMECA	DD-HAT	P	P	PD-DD	Qual./Quant.	SS, Engr., M&S	In-depth	I
FaHA	DD-HAT	Y	P	PD-DD	Qual.	SS, Engr., M&S	In-depth	I
FuHA	SD-HAT, DD-HAT	Y	P	CD-PD-DD	Qual.	SS, Engr., M&S	Moderate to in-depth	I
SCA	SD-HAT, DD-HAT	P	Y	DD	Qual.	SS, Engr., M&S	Moderate to in-depth	D
PNA	SD-HAT, DD-HAT	P	N	PD-DD	Qual./Quant.	SS, Engr., M&S	In-depth	D
MA	SD-HAT, DD-HAT	P	N	PD-DD	Qual./Quant.	SS, Engr., M&S	Moderate to in-depth	D
BA	SD-HAT	Y	P	PD-DD	Qual.	SS, Eg	Moderate to in-depth	I
BPA	DD-HAT	Y	P	PD-DD	Qual.	SS, Engr., M&S	In-depth	D
HAZOP	SD-HAT, DD-HAT	Y	P	PD-DD	Qual.	SS, Engr., M&S	Moderate to in-depth	I
CCA	SD-HAT, DD-HAT	Y	P	PD-DD	Qual./Quant.	SS, Engr., M&S	Moderate to in-depth	D
CCFA	SD-HAT, DD-HAT	Y	P	PD-DD	Qual.	SS, Engr., M&S	Moderate to in-depth	D
MORT	SD-HAT, DD-HAT	Y	P	PD-DD	Qual./Quant.	SS, M&S	Moderate to in-depth	D
SWSA	SD-HAT, DD-HAT	Y	P	CD-PD	Qual.	SS, Engr., M&S	Moderate to in-depth	N/A

Inductive Vs Deductive Techniques

	Inductive	Deductive
Methodology	✓ Going from the specific to the general	✓ Going from the general to the specific
General characteristics	✓ System is broken down into individual components ✓ Potential failures for each component ✓ Effects of each failure ✓ What-if?	✓ General nature of the hazard has already been identified ✓ System is reviewed to define the cause of each hazard ✓ How-can?
Applicability	✓ Systems with few components ✓ Systems where single-point failures (SPFs) are predominant ✓ Preliminary or overview analysis	✓ All sizes of systems ✓ Developed for complex systems ✓ Designed to identify hazards caused by multiple failures
Potential pitfalls	✓ Difficult to apply to complex systems ✓ Large number of components to consider ✓ Consideration of failure combinations becomes difficult	✓ Detailed system documentation required ✓ Large amount of data involved ✓ Time consuming

Which is better?

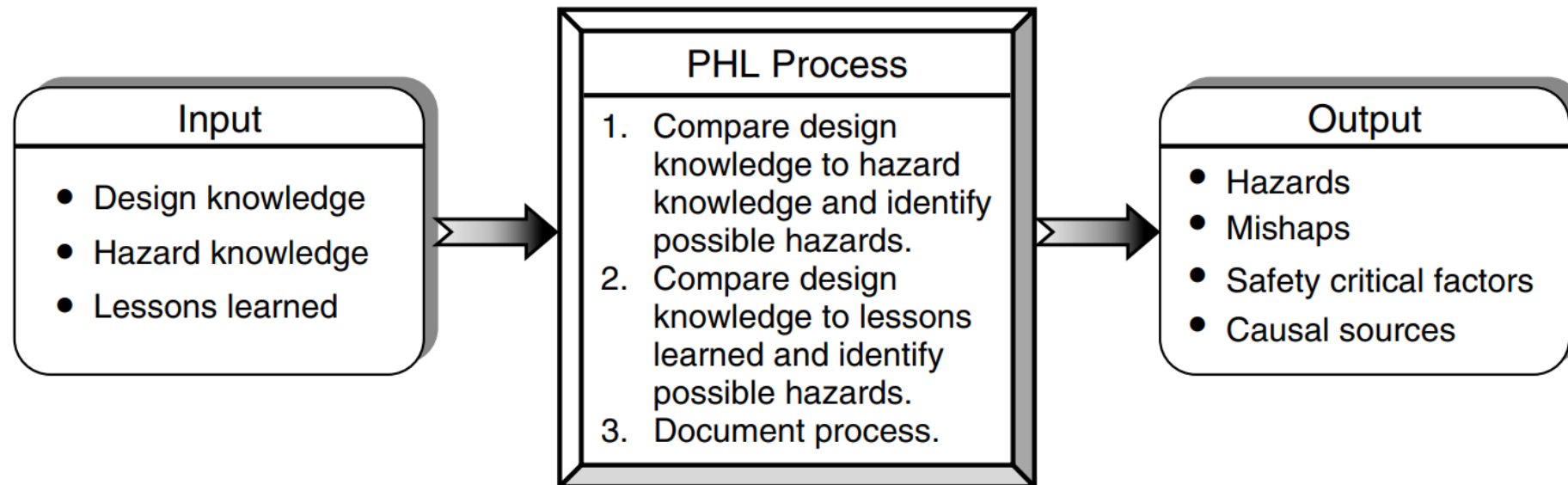
Qualitative Vs Quantitative

	Qualitative	Quantitative
Numerical results	No	Yes
Cost	Lower	Higher
Subjective/objective	Subjective	Objective
Difficulty	Lower	Higher
Complexity	Lower	Higher
Data	Less detailed	More detailed
Technical expertise	Lower	Higher
Time required	Lower	Higher
Tools required	Seldom	Usually
Accuracy	Lower	Higher

Which is better?

Preliminary Hazard List

- ✓ During conceptual or preliminary design - starting point for all subsequent hazard analyses
- ✓ Every hazard identified on the PHL will be analyzed with more detailed analysis techniques
- ✓ Primary purpose?
- ✓ Secondary purpose: to identify safety critical parameters & mishap categories
- ✓ Intent: to affect the design for safety ASAP in the development program
- ✓ Can be performed on a subsystem, a single system, or an integrated set of systems

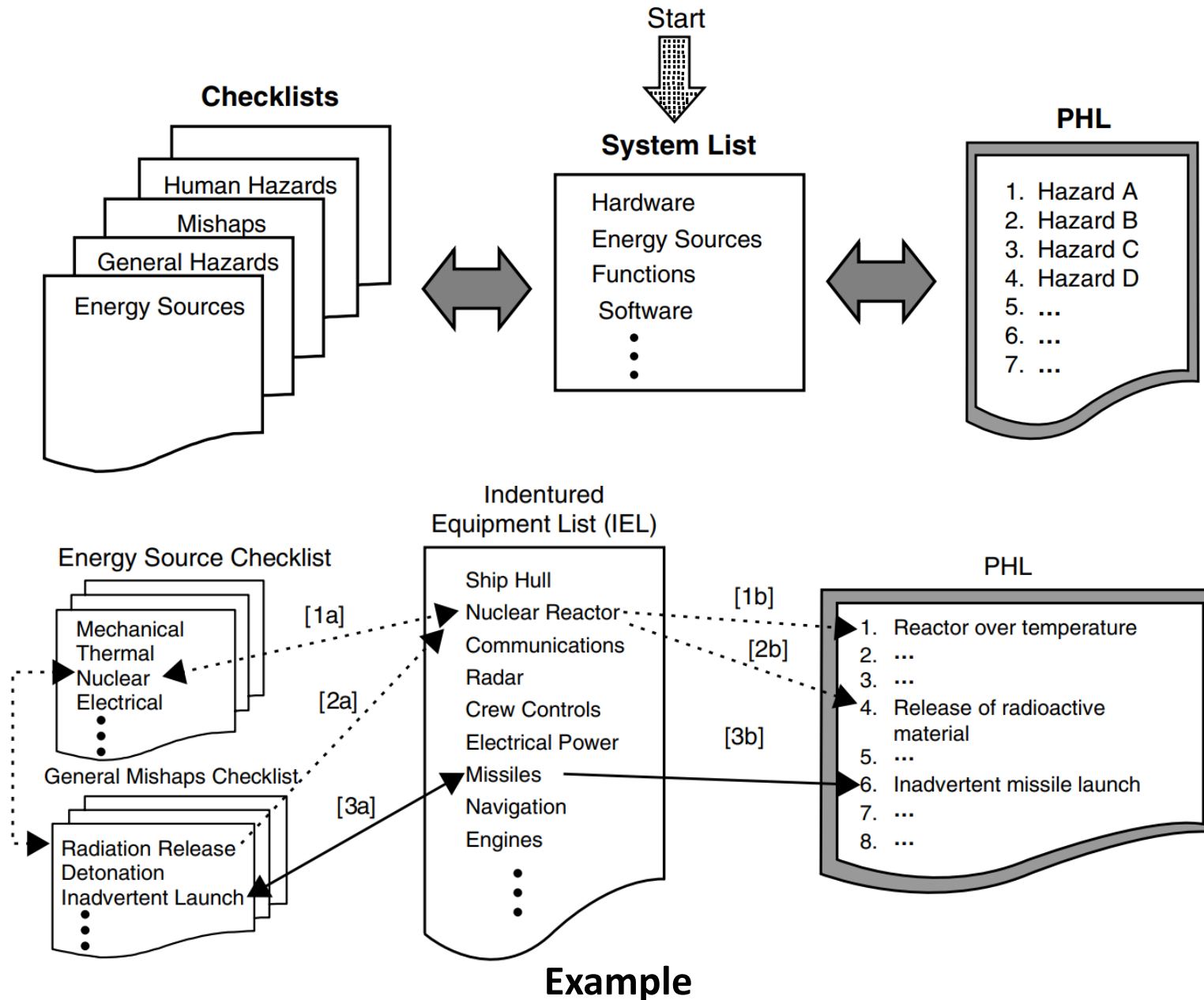


Overview of PHL

Steps of PHL

1. **Define system:** Define, scope, & bound the system. Define the mission, mission phases, & mission environments. Understand the system design, operational concepts, & major system components
2. **Plan PHL:** Establish PHL goals, definitions, worksheets, schedule, & process. Identify system elements & functions to be analyzed
3. **Select team:** Select all team members to participate in PHL & establish responsibilities. Utilize team member expertise from several different disciplines (*e.g.* design, test, manufacturing, *etc.*)
4. **Acquire data:** Acquire all of the necessary design, operational, & process data needed for the analysis (*e.g.* equipment lists, functional diagrams, operational concepts, *etc.*). **Acquire hazard checklists**, lessons learned, & other hazard data applicable to the system.
5. **Conduct PHL:** Construct list of hardware components & system functions. Evaluate conceptual system hardware, system operational functions, system energy sources, system software functions. **Compare with hazard checklists**
6. **Build hazard list:** Develop list of identified & suspected system hazards & potential system mishaps. Identify SCFs & TLMs if possible from information available
7. **Recommend corrective action:** Recommend safety guidelines & design safety methods that will eliminate or mitigate hazards
8. **Document PHL:** Document the entire PHL process & PHL worksheets in a PHL report. Include conclusions & recommendations
 - **Typical hazard checklists**
 - ✓ Energy sources, Hazardous functions, Hazardous operations, Hazardous components, Hazardous materials, Lessons learned from similar type systems, Undesired mishaps, Failure mode & failure state considerations

PHL Methodology



PHL worksheet

Preliminary Hazard List Analysis				
System Element Type: 1				
No.	System Item	Hazard	Hazard Effects	Comments
2	3	4	5	6

Hazard checklist for energy sources

- 1. Fuels**
- 2. Electrical generators**
- 3. Propellants**
- 4. RF energy sources**
- 5. Radioactive energy sources**
- 6. Explosive charges**
- 7. Falling objects**
- 8. Charged electrical capacitors**
- 9. Catapulted objects**
- 10. Storage batteries**
- 11. Heating devices**
- 12. Static electrical charges**
- 13. Pumps, blowers, fans**
- 14. Pressure containers**
- 15. Rotating machinery**
- 16. Spring-loaded devices**
- 17. Actuating devices**
- 18. Suspension systems**
- 19. Nuclear**
- 20. Gas generators**
- 21. Cryogenics**

Hazard checklist for general sources

1. Acceleration/deceleration
2. Falling objects
3. Fragments/projectiles
4. Impacts
5. Inadvertent motion
6. Sloshing liquids
7. Contamination
8. Corrosion
9. Chemical dissociation
10. Chemical replacement/combination
11. Corrosion due to electrolysis
12. Hydrogen embrittlement
13. Moisture
14. Oxidation
15. Organic (fungal/bacterial, etc.)
16. Particulate
17. Stress corrosion
18. Electrical
19. Arcing
20. Bent pins
21. Breakdown of dielectric
22. Corona
23. Distribution feedback
24. Electrical noise
25. Electrical surges
26. Electromagnetic interference
27. Excessive solder
28. Burns
29. Grounding
30. Improper electrical connections & wiring
31. Inadequate heat dissipation
32. Inadvertent activation
33. Incorrect voltage, current, cycle, etc.
34. Inductive or capacitive coupling
35. Lightning strike
36. Magnetic surge
37. Mis-mating of power connectors
38. Polarity
39. Poor insulation
40. Power outage/Power source failure
41. Shock
42. Short, open circuit
43. Static discharge
44. Stray currents/sparks

Hazard checklist for general sources (contd..)

1. Control systems

- 2. Inappropriate operation
- 3. Inappropriate software operation
- 4. Interference to control system
- 5. Sneak circuit

6. Weather & environment

- 7. Fog
- 8. Foreign matter contamination
- 9. Fungal/bacterial
- 10. High humidity
- 11. Low humidity
- 12. Lightning
- 13. Outside versus inside environment
- 14. Precipitation (rain, snow, ice, sleet, hail, etc.)
- 15. Radiation
- 16. Salt
- 17. Sand/dust
- 18. Temperature extremes (& variation)
- 19. Vacuum
- 20. Wind

21. Ergonomic

22. Fatigue

23. Faulty/inadequate control/readout labeling

24. Faulty workstation design

25. Glare

26. HVAC

27. Inaccessibility

28. Inadequate/improper illumination

29. Explosion/explosive

30. Chemical contamination

31. Dust explosion

32. Electrostatic discharge

33. Friction

34. Impact/shock

35. Lightning

36. Pyrophoric

37. Fire

38. Chemical change (exothermic/endothermic)

39. Combustible material and combustible atmosphere

40. Fuel, oxidizer, & ignition source

Hazard checklist for general sources (contd..)

1. **Human factors**
2. Failure to operate
3. Inadvertent operation
4. Operation too brief/too long
5. Operation early/late
6. Operation out of sequence
7. Leaks/spills
8. Dusts/solids
9. Flooding
10. Gases/vapors
11. Liquids
12. Porosity
13. Radioactive leaks
14. Runoff
15. **Life cycle**
16. Maintenance
17. Start-up
18. Steady-state operation
19. Stressed operation
20. Shutdown (standard, emergency, unexpected)
21. **Materials**
22. Chemical combinations
23. Compressible/incompressible fluids
24. Combustible material
25. Dissimilar materials
26. Halogens & other oxidizing agents
27. Lack of resiliency
28. Lubrication
29. Incompatible materials/media
30. Solvent residues
31. Toxicity
32. **Mechanical**
33. Thermal Shock
34. High/Low temperature
35. Vibration & noise
36. Pressure
 - High
 - Low
 - Rapid change
37. Crushing surfaces
38. Ejected parts/fragments
39. Fatigue/cyclic stresses
- 40.

Hazard checklist for general sources (contd..)

1. Flexure
2. Shock (mechanical)
3. Friction surfaces
4. Hysteresis
5. Lifting
6. Misalignment
7. Pinch points
8. Rotating equipment
9. Sharp edges
10. Stability/toppling potential
11. Torquing (over/under)
12. Stress concentrations
13. Stress reversals
14. Structural damage or failure
15. Gravity
16. Collision
17. Acceleration/deceleration
18. Bad welds
19. Cracks
20. Brittleness/ductility of materials
21. Physiological
22. Allergens
23. Asphyxiates
24. Carcinogens
25. Irritants
26. Mutagens
27. Vibration & noise
28. Pathogens
29. Radiation
30. Radiation
31. Ionizing (γ , β , α , X-ray)
32. Non-ionizing (infrared, laser, microwave, ultraviolet)
33. Thermal
34. Temperature
35. Altered structural properties
36. Burns
37. Compressive heating
38. Cryogenic properties
39. Solar effects
40. Joule-Thomson cooling

Hazard checklist for general sources (contd..)

1. Pneumatic/hydraulic pressure/vacuum
2. Backflow/siphon effect
3. Blast
4. Cavitation
5. Hydraulic hammer
6. Implosion
7. Inadequate pressure
8. Inadequate release of material
9. Over/under pressurization
10. Pipe/vessel rupture
11. Pipe/vessel whip
12. Rapid pressure change
13. Cyber controlled
14. Loss of capability

Hazard checklist for general operations

1. Welding
2. Cleaning
3. Extreme temperature operations
4. Extreme weight operations
5. Hoisting, handling, & assembly operations
6. Test chamber operations
7. Proof test of major components/subsystems/systems
8. Propellant loading/transfer/handling
9. High-energy pressurization/hydrostatic-pneumatic testing
10. Nuclear component handling/checkout
11. Ordnance installation/checkout/test
12. Tank entry/confined space entry
13. Transport & handling of end item
14. Manned vehicle tests
15. Static firing

Hazard checklist for failure states

1. Fails to operate
2. Operates incorrectly/erroneously
3. Operates inadvertently
4. Operates at incorrect time (early, late)
5. Unable to stop operation
6. Receives erroneous data
7. Sends erroneous data

Ace missile system – System hardware category

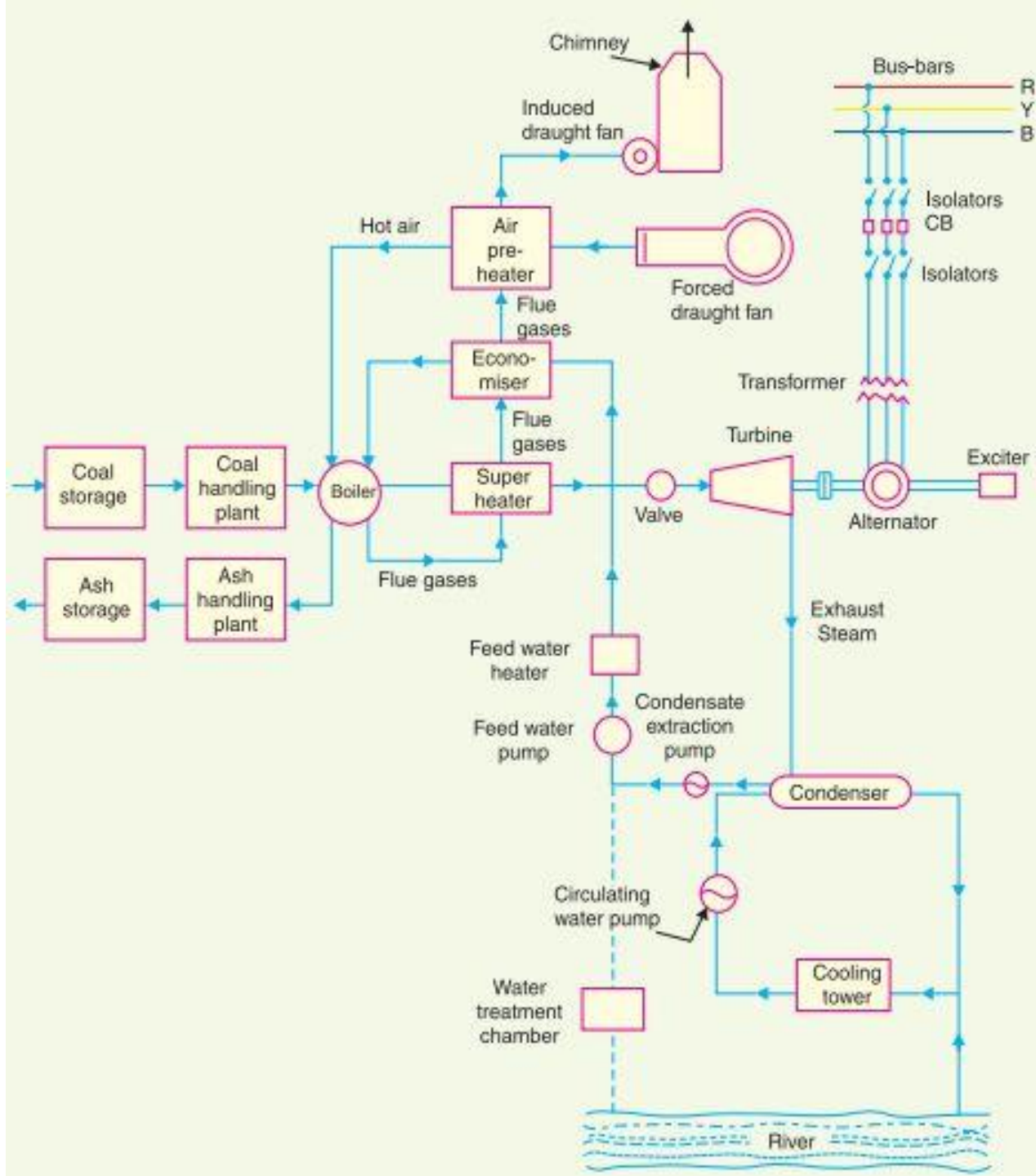
Start with first item in system hardware category

PHL				
No.	Item	Hazard	Effect	Comments
PHL-1	Missile	Inadvertent missile launch	Unintended launch; crash	
PHL-2				
PHL-3				

Look for “Missile” in hazard checklist. Find “Inadvertent Launch” as a potential hazard. Note simplified hazard write-up.

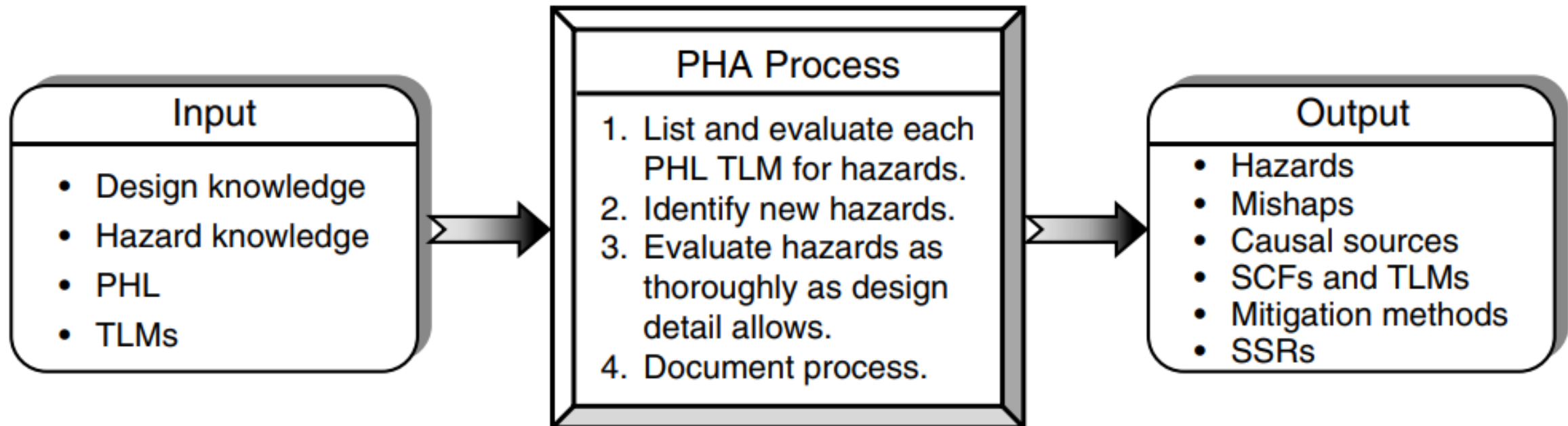
State system effect for hazard.

Thermal Power Plant



Preliminary Hazard Analysis

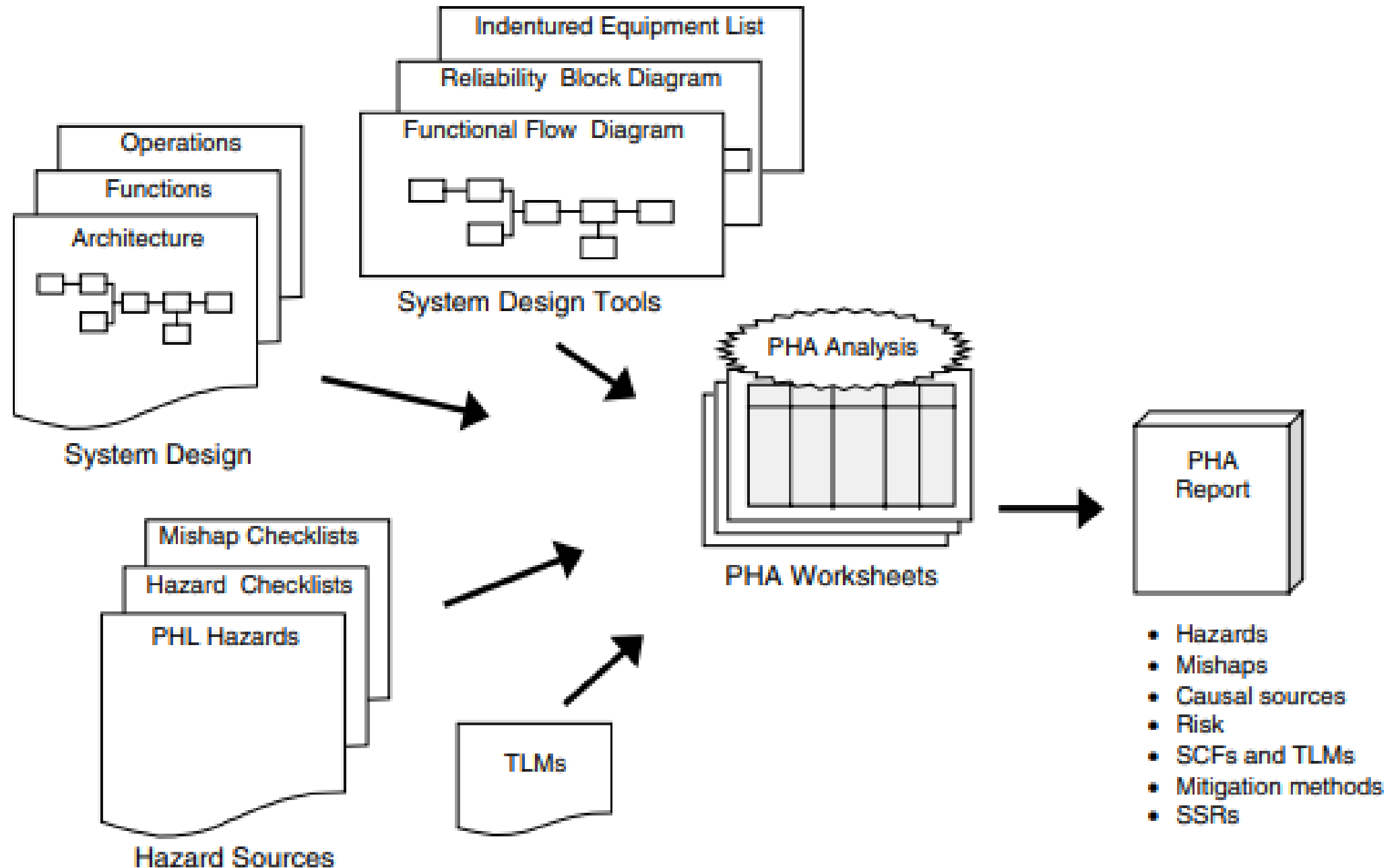
- ✓ Falls under PD-HAT type
- ✓ Identifying hazards, their associated causal factors, effects, level of risk, & mitigating design measures when detailed design information is not available
- ✓ Every hazard identified on the PHL will be analyzed with more detailed analysis
- ✓ Gross hazard analysis & potential hazard analysis
- ✓ Intent: to affect the design for safety ASAP in the development program
- ✓ Can be performed on a subsystem, a single system, or an integrated set of systems



Steps of PHA

- 1. Define system:** Define, scope, & bound the system. Define the mission, mission phases, & mission environments. Understand the system design, operational concepts, & major system components
- 2. Plan PHL:** Establish PHA definitions, worksheets, schedule, & process. Identify system elements & functions to be analyzed
- 3. Establish safety criteria:** Identify applicable design safety criteria, safety precepts/principles, safety guidelines, & safety critical factors
- 4. Acquire data:** Acquire all of the necessary design, operational, & process data needed for the analysis (e.g., functional diagrams, **drawings**, operational concepts, etc.). Acquire hazard checklists, lessons learned, & other hazard data applicable to the system. Acquire all **regulatory data** & information that are applicable.
- 5. Conduct PHA:** Construct list of equipment, functions, & energy sources. Compare conceptual system hardware, system operational functions, system energy sources, system software functions with hazard checklists & TLMs. **Expand the list of SCFs & TLMs & utilize in the analysis. Be cognizant of functional relationships, timing, & concurrent functions when identifying hazards.**
- 6. Evaluate risk:** Identify the level of mishap risk presented for each identified hazard, both with & without hazard mitigations in the system design
- 7. Recommend corrective action:** Recommend safety guidelines & design safety methods that will eliminate or mitigate hazards. **Translate the recommendations into SSRs. Also, identify safety features already in the design or procedures that are present for hazard mitigation.**
- 8. Monitor corrective action:** Review test results to ensure that safety recommendations & SSRs are effective in mitigating hazards as anticipated
- 9. Track hazards:** Transfer newly identified hazards into the HTS. Update the HTS as hazards, HCFs, & risk are identified in the PHA
- 10. Document PHA:** Document the entire PHL process & PHL worksheets in a PHL report. Include conclusions & recommendations

PHA Methodology



PHA worksheet

System: 1		Preliminary Hazard Analysis					Analyst: 3		
Subsystem/Function: 2		Date: 4							
No.	Hazard	Causes	Effects	Mode	IMRI	Recommended Action	FMRI	Comments	Status
5	6	7	8	9	10	11	12	13	14
			Severity		Probability				
			I. Catastrophic		A. Frequent				
			II. Critical		B. Probable				
			III. Marginal		C. Occasional				
			IV. Negligible		D. Remote				
					E. Improbable				

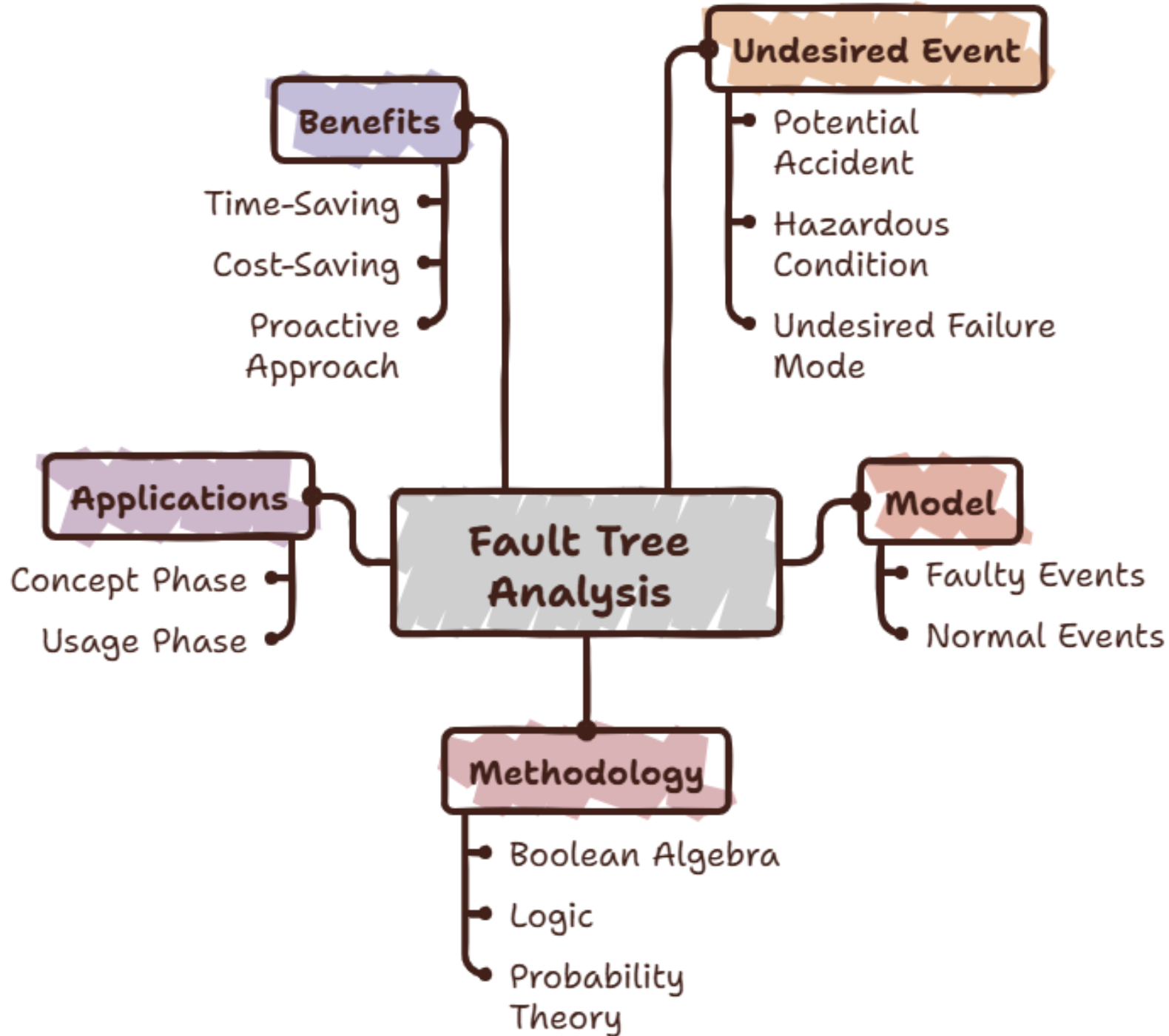
PHA worksheet

System: Ace Missile System Subsystem: Missile Structure Subsystem			Preliminary Hazard Analysis				Analyst: Date:		
No.	Hazard	Causes	Effects	Mode	IMRI	Recommended Action	FMRI	Comments	Status
PHA-1	Missile structure fails, resulting in unstable missile flight and missile crash	Manufacturing defect; design error	Unstable flight, resulting in crash causing death/injury; incorrect target	Flight	1D	Use 5× safety factor on structure design	1E		Open
PHA-2	Missile body breaks up, resulting in fuel leakage; and ignition source, causing fire	Manufacturing defect; design error	Missile fire, causing death/injury	Ground operations	1D	Use 5× safety factor on structure design	1E		Open
PHA-3	Missile structure fails during handling, resulting in personnel injury	Manufacturing defect; design error; handling equipment failure	Personnel injury	PHS&T ^a	2D	Use 5× safety factor on structure design Establish SSRs for handling equipment	2E		Open

Fault tree analysis

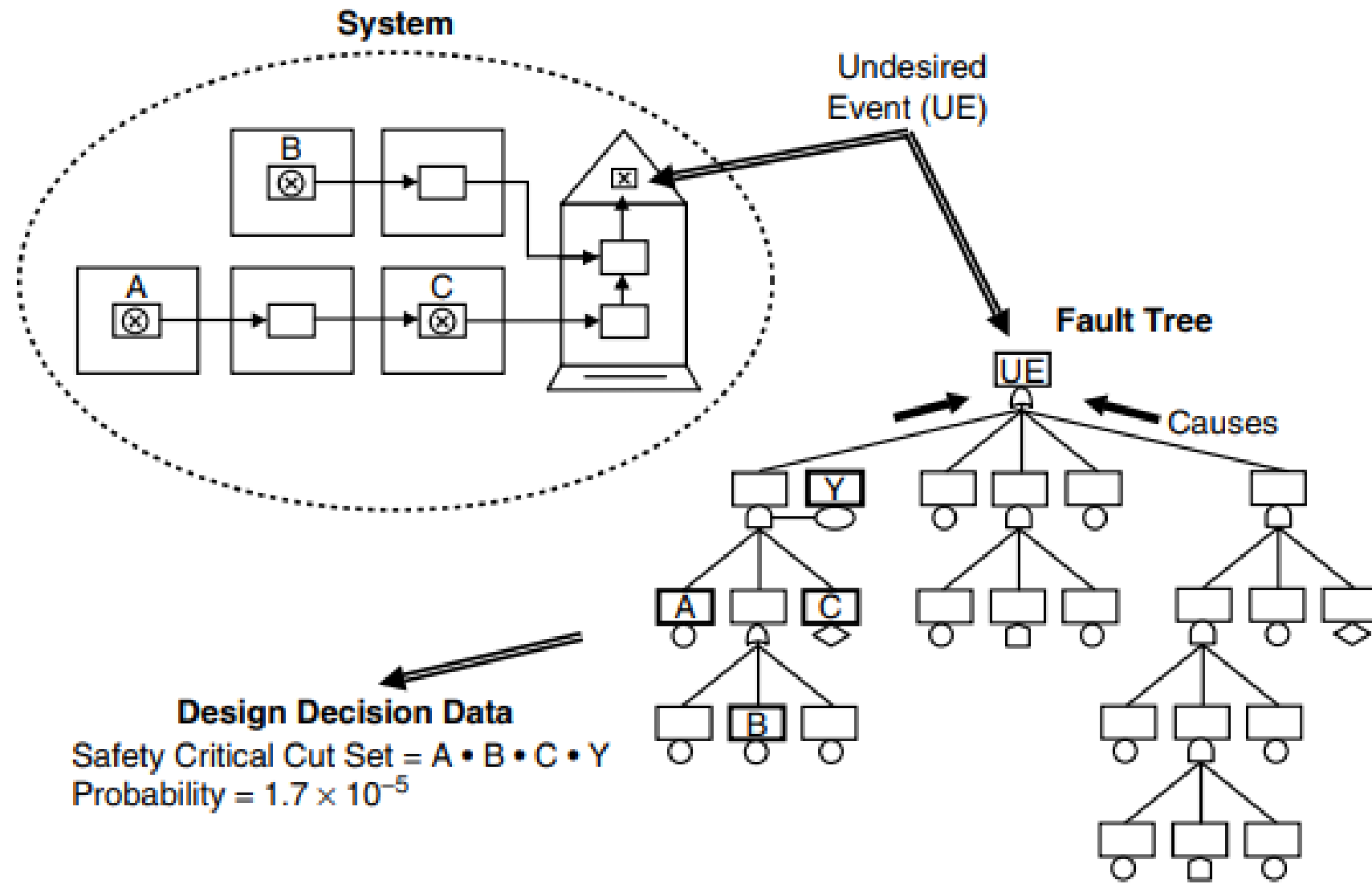
1. Systems analysis technique used to determine the root causes & probability of occurrence of a specified undesired event
2. An undesired event: event that is identified as objectionable & unwanted
3. Such as a potential accident, hazardous condition, or undesired failure mode
4. I/D?
5. Qualitative/quantitative
6. A model that logically & graphically represents the various combinations of possible events, both faulty & normal, occurring in a system that lead to an UE
7. AKA logic tree analysis & logic diagram analysis
8. Falls under SD-HAT, can be applied during any lifecycle phase of a system—from concept to usage – Earlier is better
9. Time- & cost-saving feature: only those system elements that contribute to the occurrence of the undesired event need to be analyzed
10. Robust, rigorous, & structured methodology
11. Although classified as a hazard analysis, it is primarily used as a root cause analysis (RCA) tool to identify & evaluate HCFs. In addition, it can provide a probability risk assessment
12. Proactive Vs Reactive

FTA

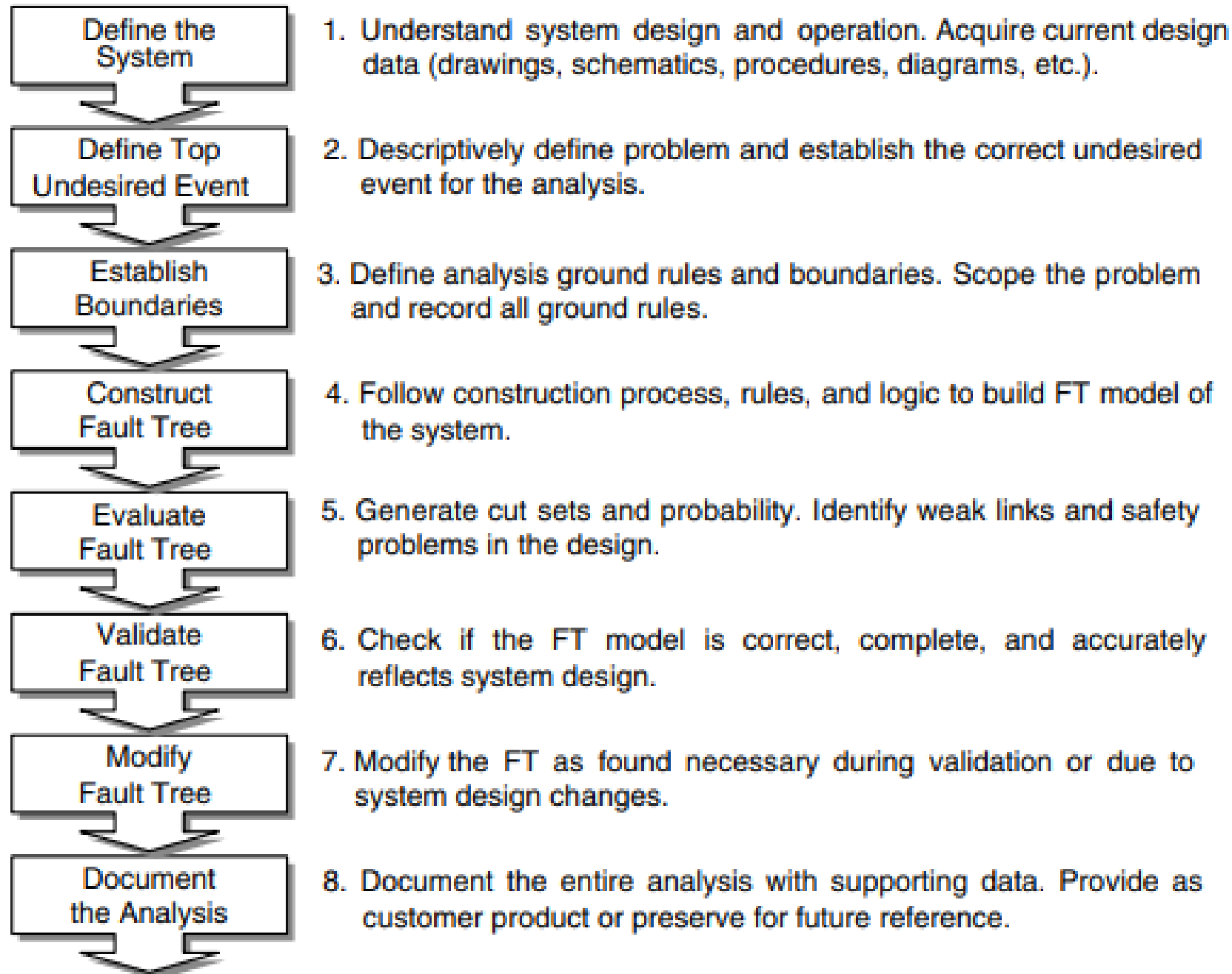


FTA Overview

1. Not suitable when single point of failures
2. The strength: ability to identify combinations of basic equipment & human failures that can lead to an accident, allowing the analyst to focus preventive measures on significant basic causes.
3. Preferred to analyze highly-redundant systems & high-energy systems with high severity



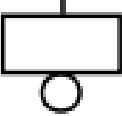
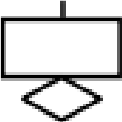
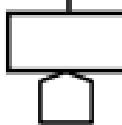
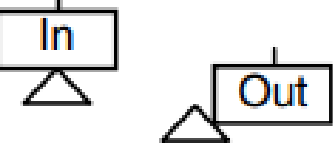
FTA Steps



Building Blocks

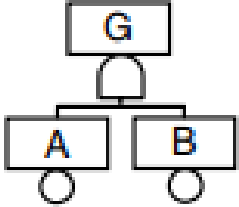
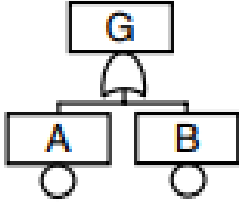
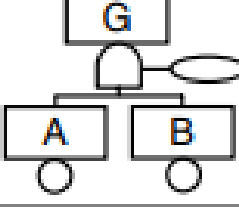
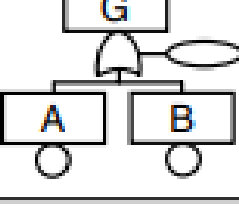
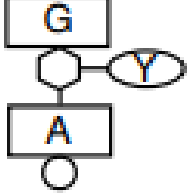
1. Categories:

- Basic events
- Gate events
- Conditional events
- Transfer events

Symbol	Type	Description
	Node Text Box	Contains the text for all FT nodes. Text goes in the box, and the node symbol goes below the box.
	Primary Failure (BE)	A basic component failure; the primary, inherent, failure mode of a component. A random failure event.
	Secondary Failure (BE)	An externally induced failure or a failure mode that could be developed in more detail if desired.
	Normal Event (BE)	An event that is expected to occur as part of normal system operation.
	Condition (CE)	A conditional restriction or probability.
	Transfer (TE)	Indicates where a branch or sub-tree is marked for the same usage elsewhere in the tree. In and Out or To/From symbols.

FT symbols for basic events, conditions, & transfers

Building Blocks

Symbol	GateType
	AND Gate
	OR Gate
	Priority AND Gate
	Exclusive OR Gate
	Inhibit Gate

The output occurs only if all of the inputs occur together

The output occurs only if at least one of the inputs occurs.

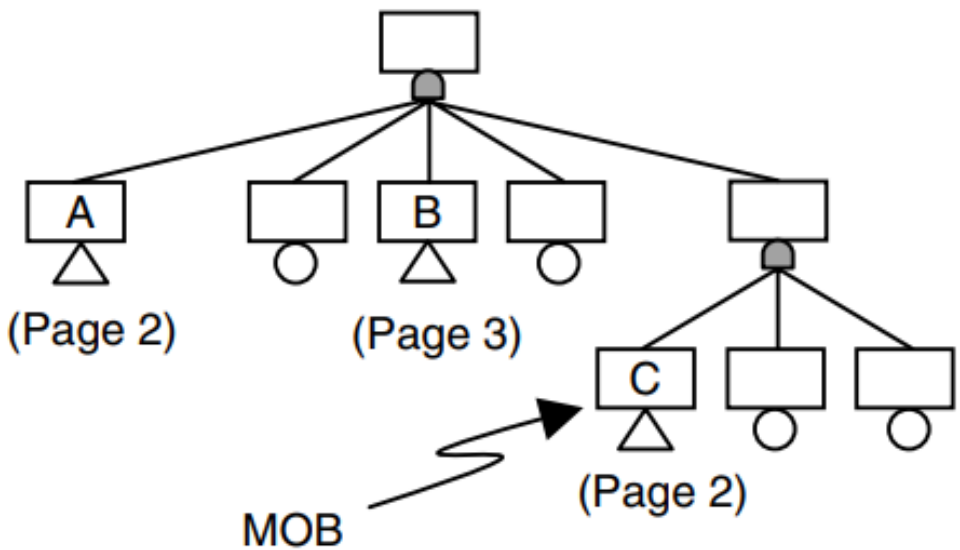
The output occurs only if all of the inputs occur together, & A must occur before B. The priority statement is contained in the Condition symbol

The output occurs if either of the inputs occurs, but not both. The exclusivity statement is contained in the Condition symbol

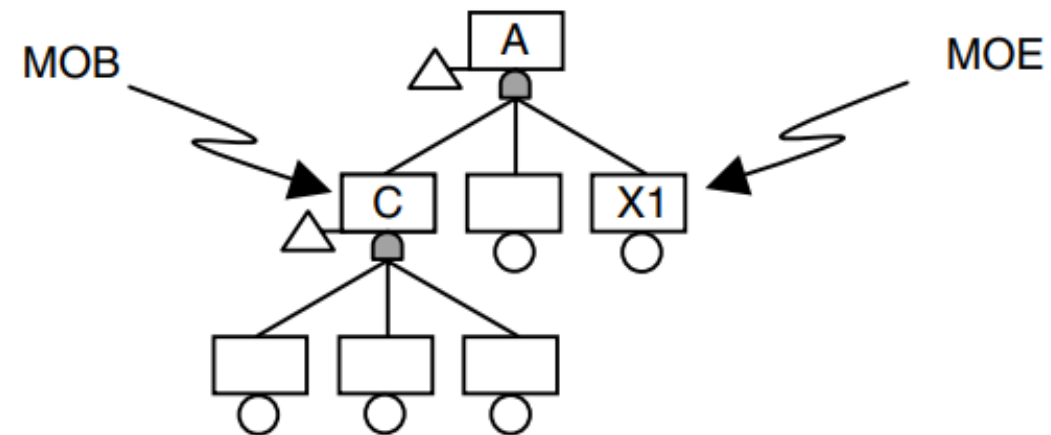
The output occurs only if the input event occurs & the attached condition is satisfied.

FT symbols for gate events

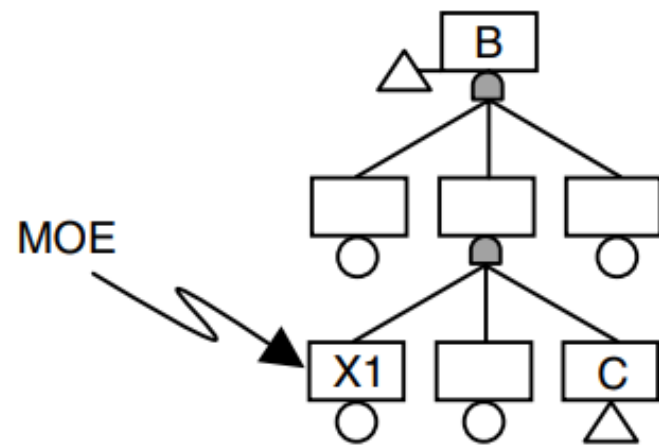
FT transfers & MOE/MOB



Page 1



Page 2

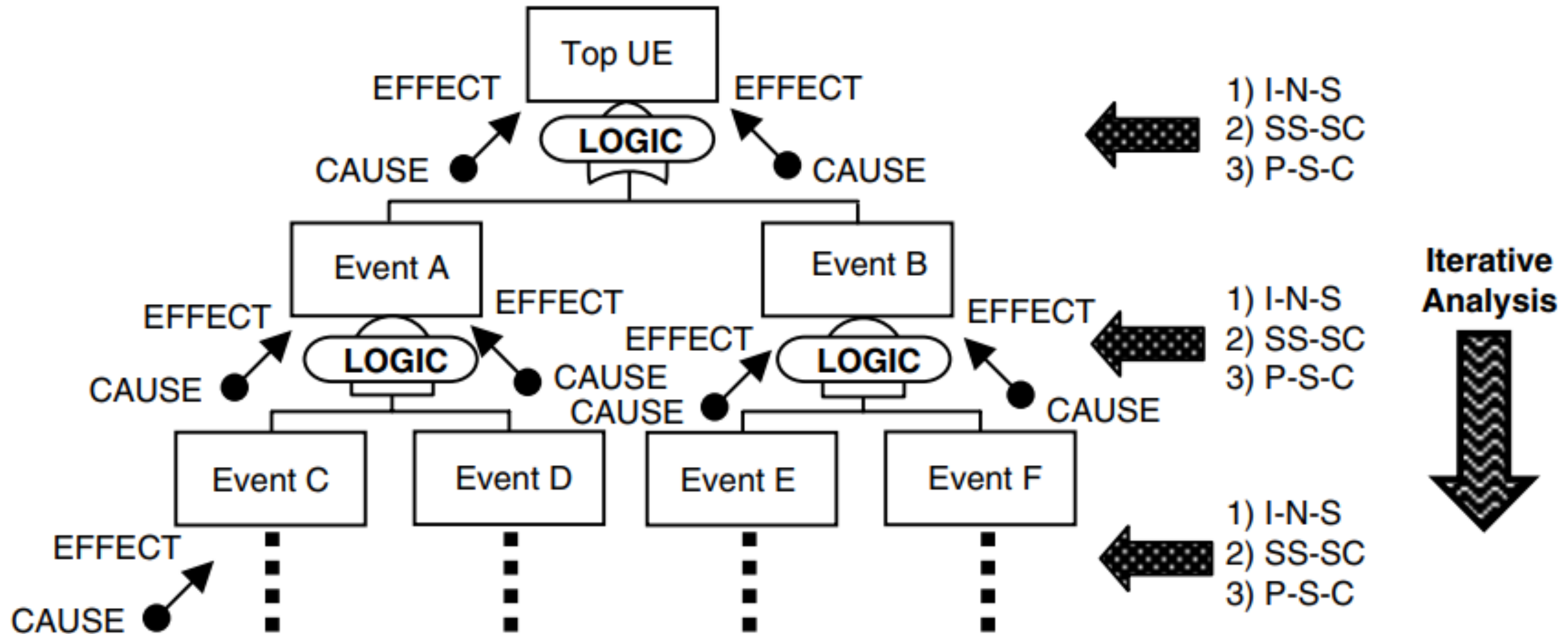


Page 3

FTA – Basics of construction

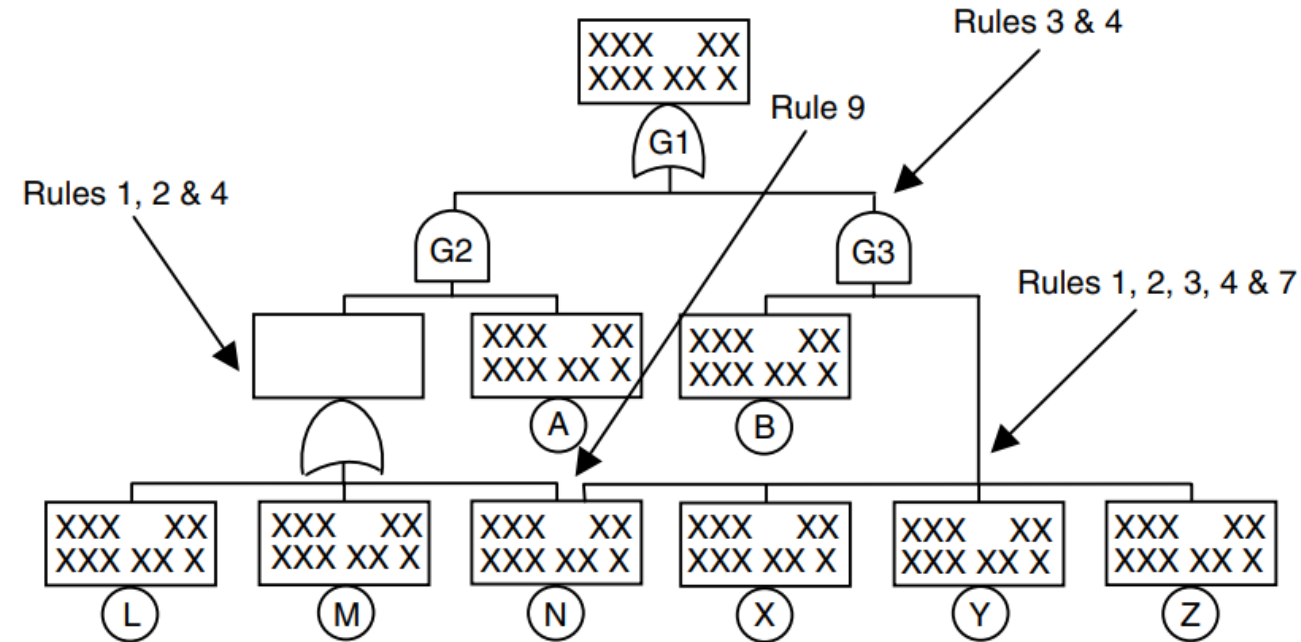
1. After identifying the top UE, sub-undesired events are identified & structured into what is referred to as the top fault tree layer. The actual deductive analysis begins with the development of the fault flow or cause-and-effect relationship of fault & normal events through the system
2. In developing the structure of the FT, certain procedures must consistently be followed in a repetitive manner
3. Procedure evolves around three principal concepts
 - The I–N–S concept
 - The SS–SC concept
 - The P–S–C concept
4. A command failure is an expected, or intended, event that occurs at an undesired time due to specific failures, e.g.: Relay coil
5. What downstream event commands the event to occur?
6. At the finish of each FT branch, the command path will terminate in primary &/or secondary events

FT building steps



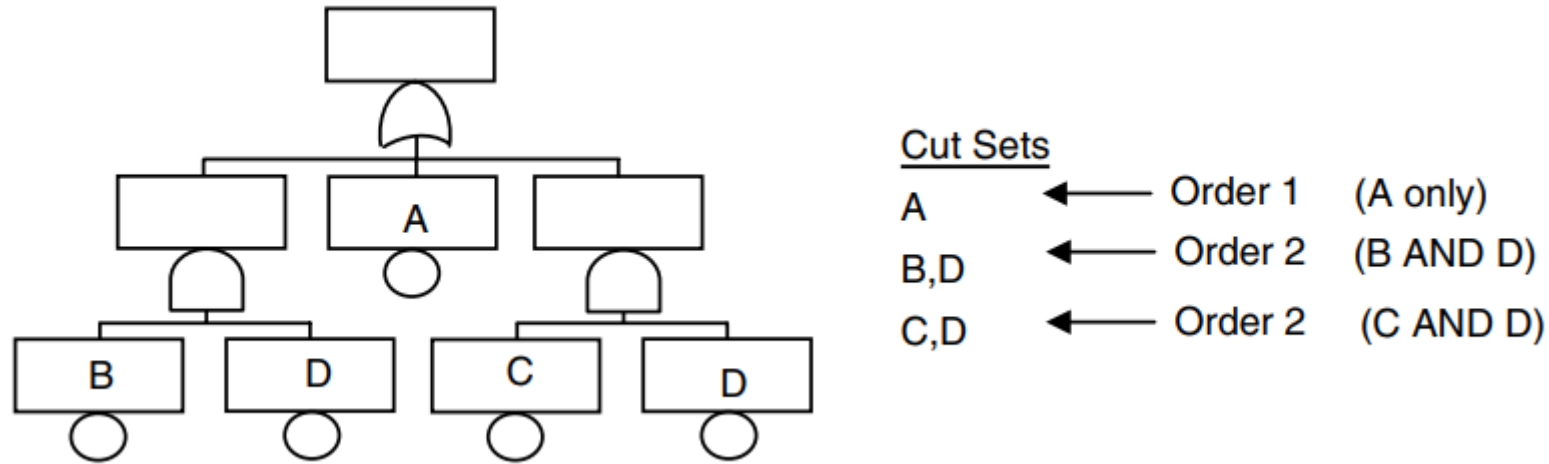
FTA – Construction errors

1. Complete basic required data for each FT node (node type, node name, & text)
2. Give every node a unique identifying name
3. No gate-to-gate connections are allowed (always have text box)
4. Always place relevant text in text box; never leave it blank
5. State event fault state exactly & precisely; use state transition wording
6. Complete the definition of all inputs to a gate before proceeding
7. Keep events on their relative level for clarity
8. Use meaningful naming convention
9. Do not draw lines from two gates to a single input (use the MOE methodology)



10. Assume no miracles (i.e., miraculous component failure blocks other failures from causing UE)
11. I–N–S, P–S–C, & SS–SC are analysis concepts; do not use these words in text boxes

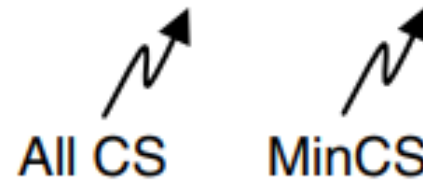
FTA – Cut sets

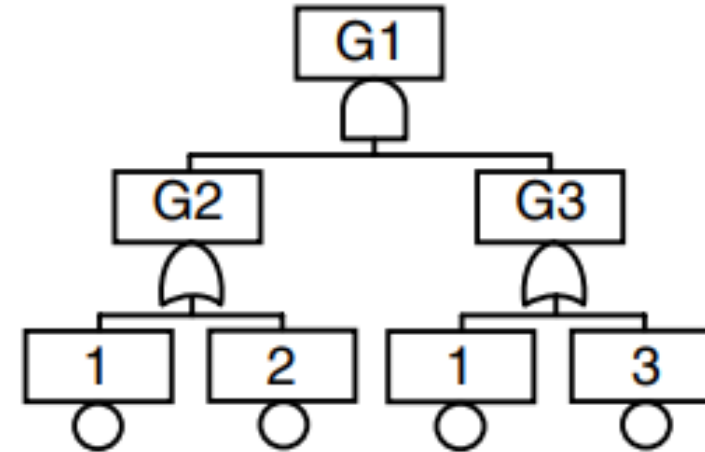


1. One of the key products from FTA
2. Identify the component failures &/or event combinations that can cause the top UE to occur
3. Provide one mechanism for probability calculations
4. A low-order CS indicates high safety vulnerability. A single-order CS (i.e., a single-point failure) tends to cause the greatest risk
5. A high-order CS indicates low safety vulnerability. A high-order CS (e.g., a five-input AND gate) tends to have a comparatively small probability & therefore presents less system risk.
6. For a large total number of CS the analyst needs to evaluate the collective risk on the top UE. This is because all of the CS added together might reach an unacceptable value.

MOCUS (method of obtaining cut sets) algorithm

1	2	3	4	5	6
G1	G2, G3	1, G3	1, 1	1	1
			1, 3	1, 3	
		2, G3	2, 1	1, 2	
			2, 3	2, 3	2, 3


 All CS MinCS



Steps:

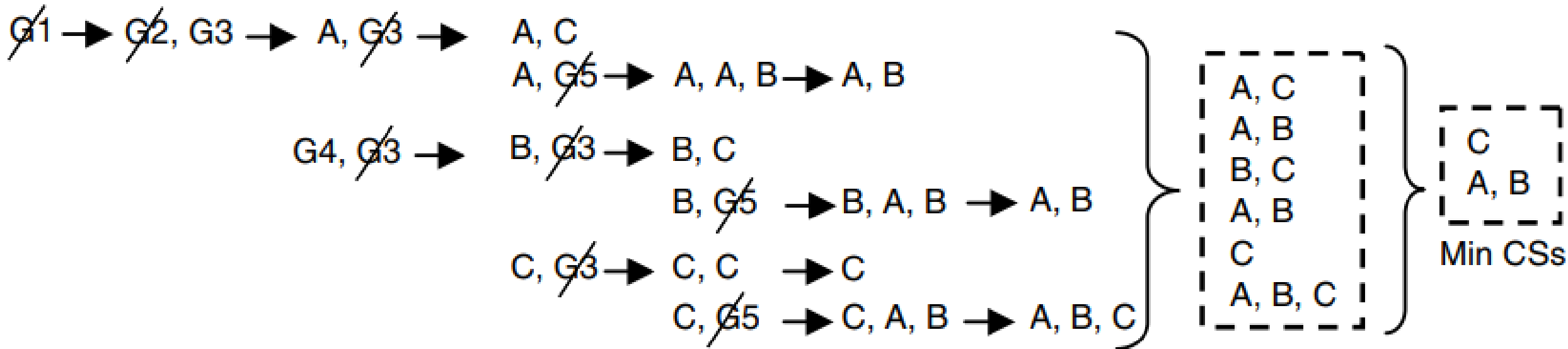
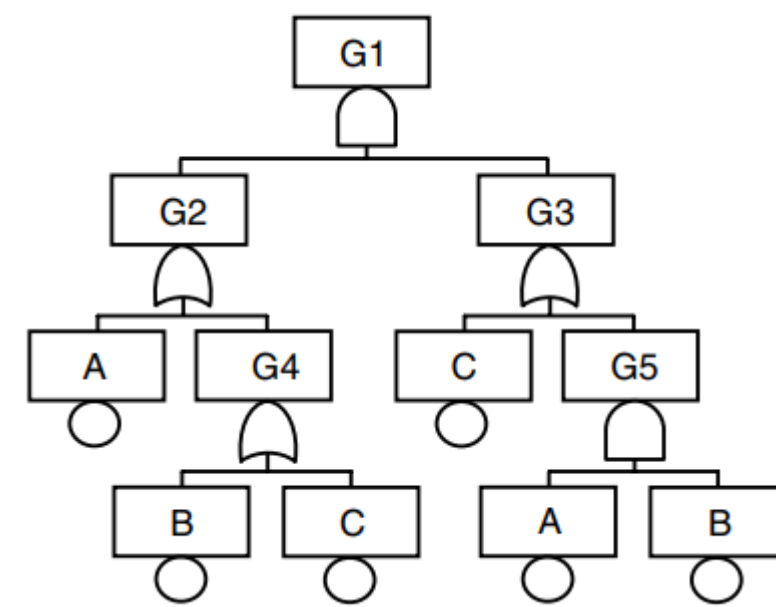
1. Enter top gate name.
2. Replace G1 with its inputs, G1 and G2.
3. Replace G2 with its inputs, 1 and 2.
4. Replace G3 with its inputs, 1 and 3.
5. These are the total CSs, some are nonminimal.
6. Eliminate nonminimal CSs.

CSs ORed

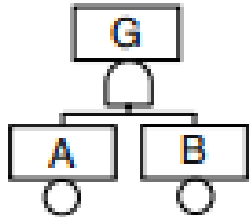
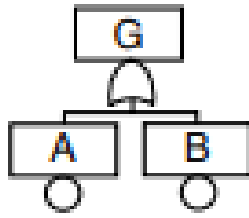
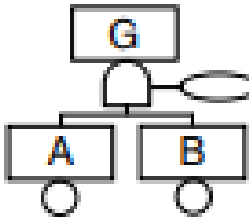
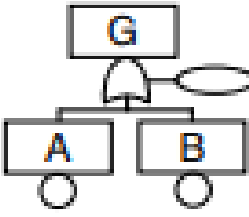
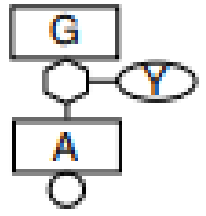
Observation: AND gates increase number of elements in a CS & that OR gates increase the number of CSs

MOCUS Vs Bottom-up Algorithm

$$\begin{aligned}
 G5 &= A \bullet B = AB \\
 G3 &= C + G5 = C + AB \\
 G4 &= B + C \\
 G2 &= A + G4 = A + B + C \\
 G1 &= G2 \bullet G3 \\
 &= (A + B + C)(C + AB) \\
 &= AC + AAB + BC + BAB + CC + CAB \\
 &= AC + AB + BC + AB + C + ABC \\
 &= C + AC + BC + AB + ABC \\
 &= C + AB
 \end{aligned}$$



Gate by gate method

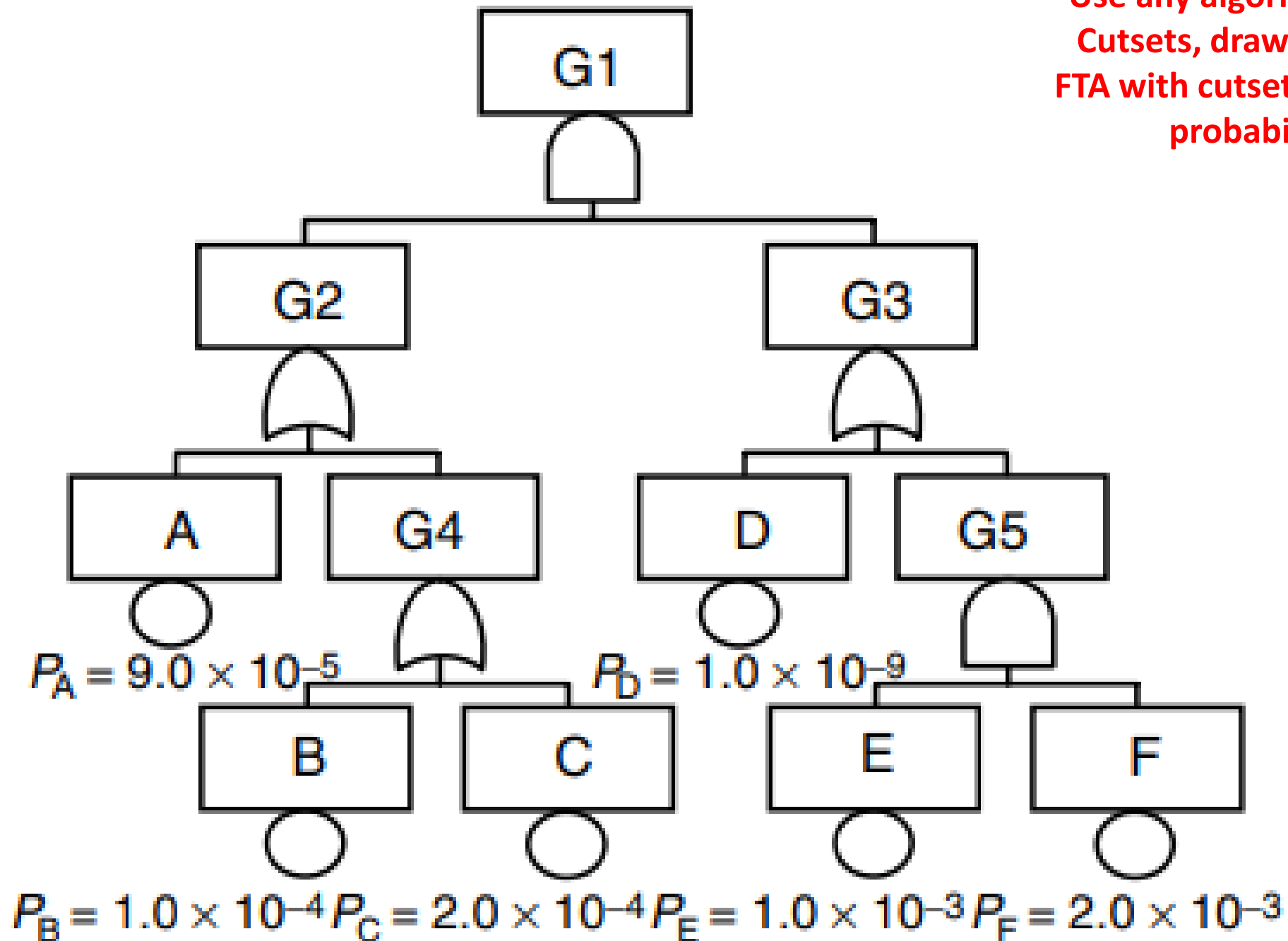
Symbol	GateType	Description
	AND Gate	The output occurs only if all of the inputs occur together. $P = P_A \cdot P_B = P_A P_B$ (2 input gate) $P = P_A \cdot P_B \cdot P_C = P_A P_B P_C$ (3 input gate)
	OR Gate	The output occurs only if at least one of the inputs occurs. $P = P_A + P_B - P_A P_B$ (2 input gate) $P = (P_A + P_B + P_C) - (P_{AB} + P_{AC} + P_{BC}) + (P_{ABC})$ (3 input gate)
	Priority AND Gate	The output occurs only if all of the inputs occur together, and A must occur before B. The priority statement is contained in the Condition symbol. $P = (P_A P_B) / N!$ Given $\lambda_A \approx \lambda_B$ and $N = \text{number of inputs to gate}$
	Exclusive OR Gate	The output occurs if either of the inputs occurs, but not both. The exclusivity statement is contained in the Condition symbol. $P = P_A + P_B - 2(P_A P_B)$
	Inhibit Gate	The output occurs only if the input event occurs and the attached condition is satisfied. $P = P_A \cdot P_Y = P_A P_Y$

Inclusion–exclusion approximation

Can you derive the expression?

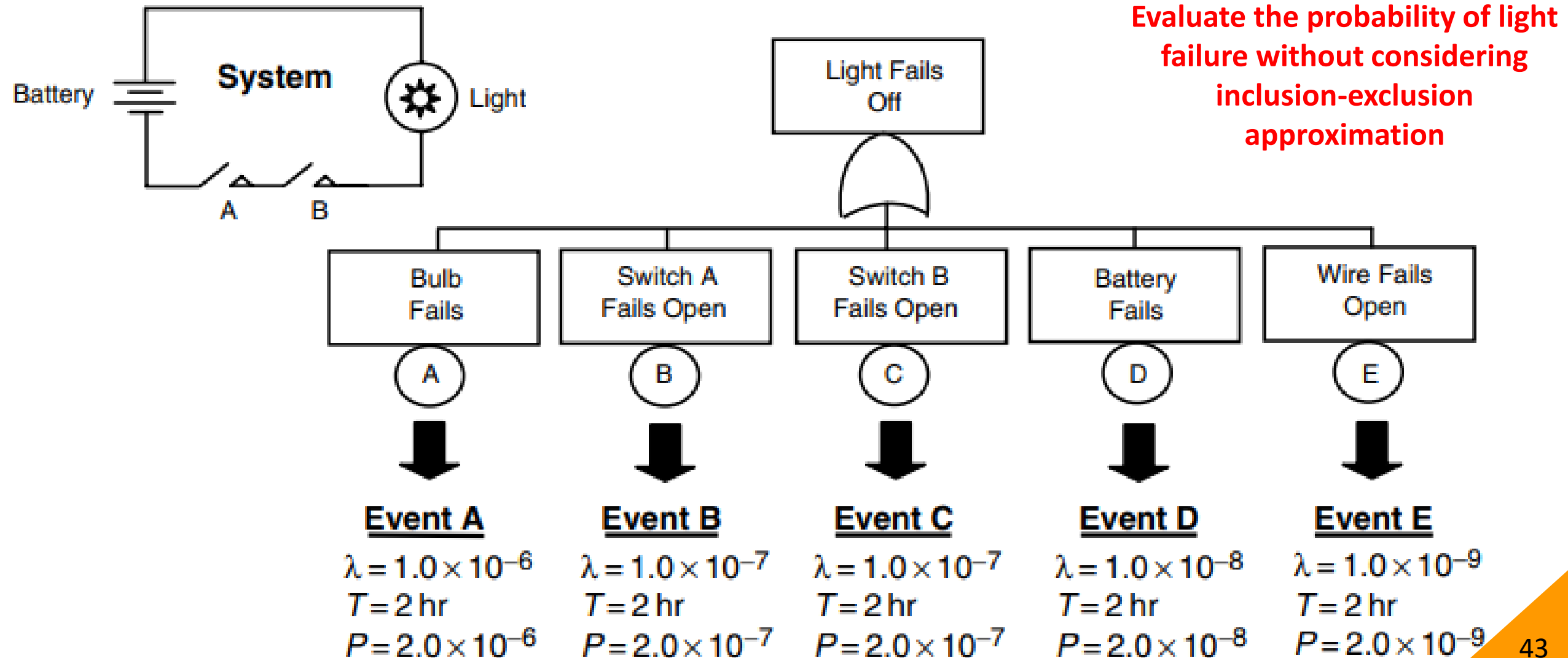
Example

Use any algorithm to arrive at Cutsets, draw the equivalent FTA with cutsets & evaluate the probability of G1

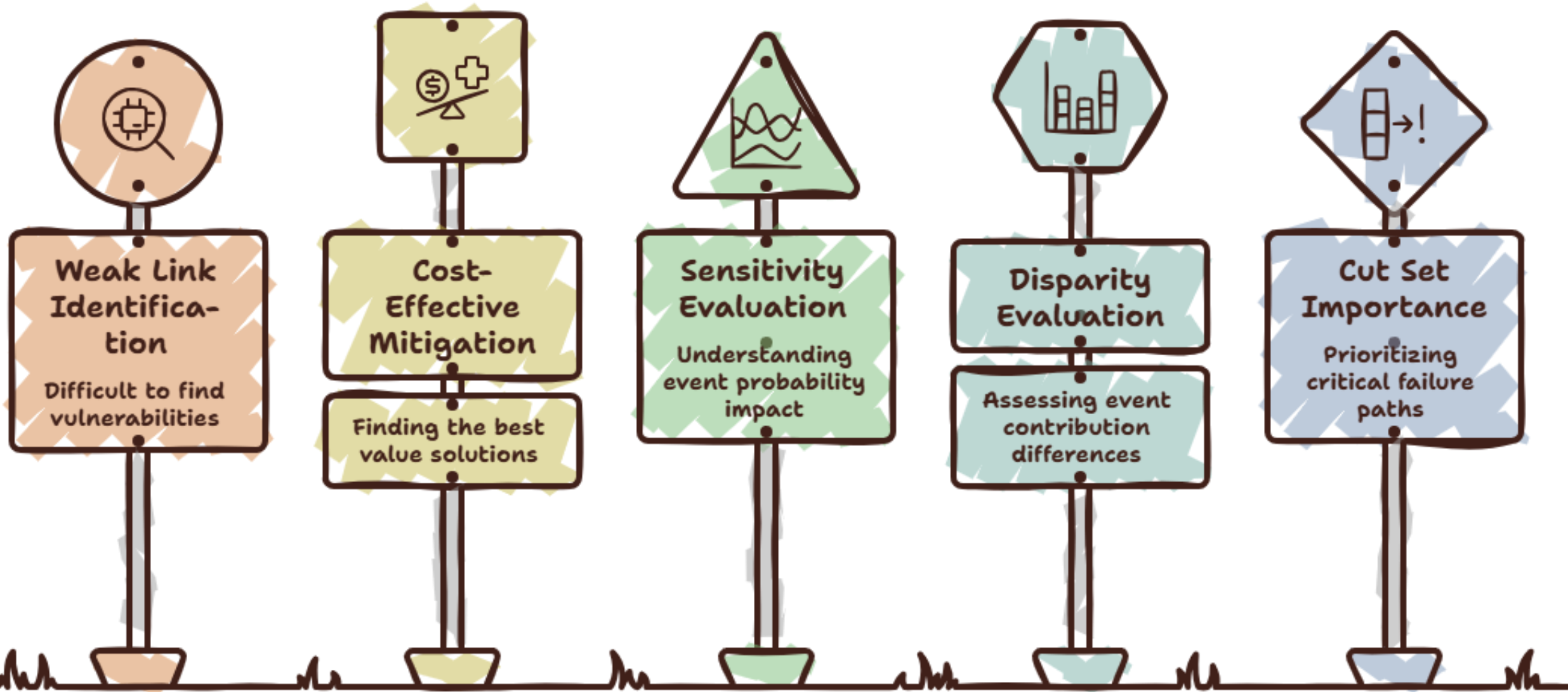


Reliability

1. Probability of success: Reliability (R) of a component, which is calculated by
2. Probability of failure: Unreliability (Q)



Importance measures

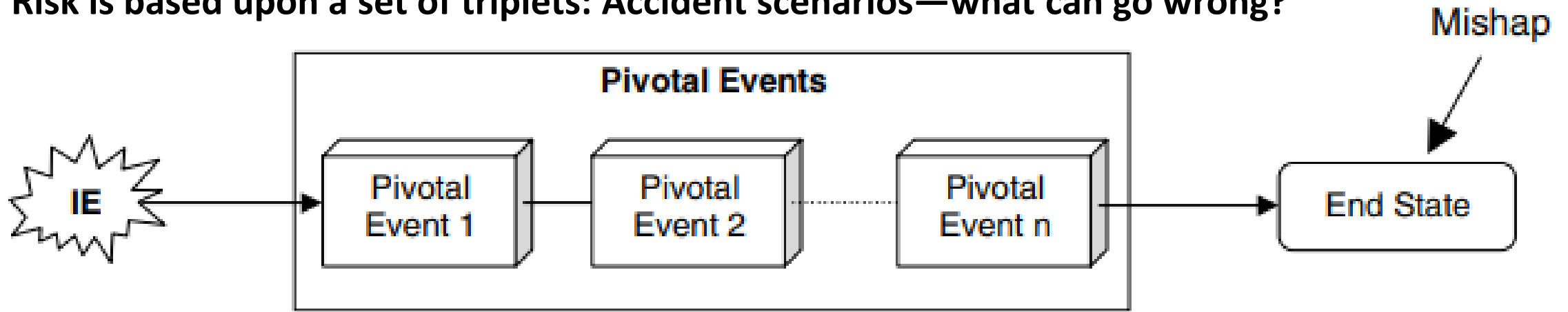


Event tree analysis (ETA)

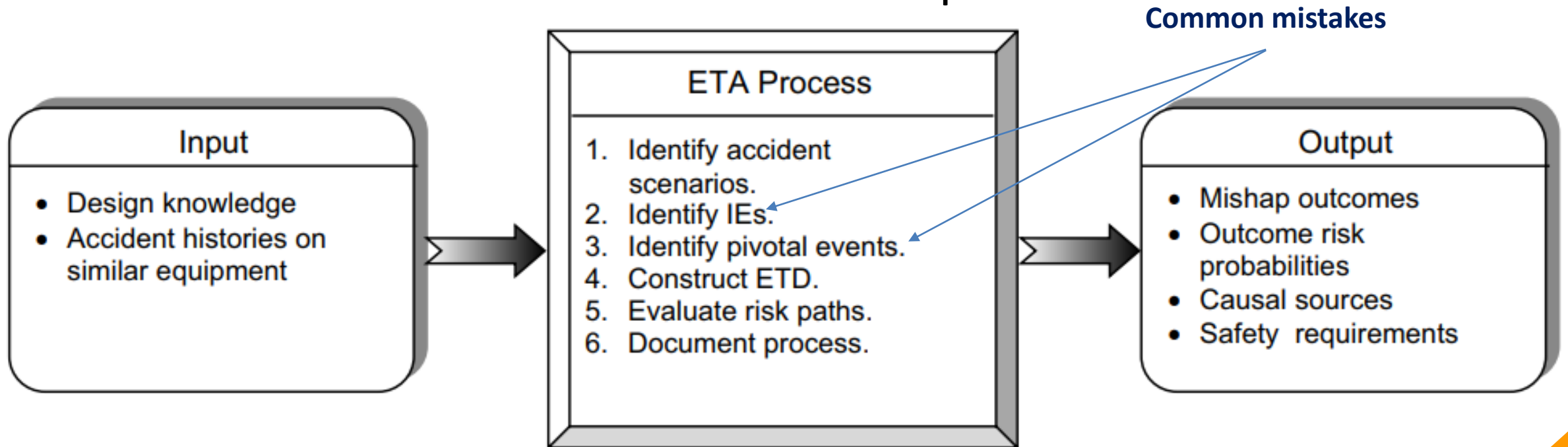
1. Alternative to FTA in 1974
2. Identifying & evaluating the sequence of events in a potential accident scenario following the occurrence of an initiating event
3. Utilizes a visual logic tree structure known as an event tree (ET) or ETD
4. Objective : to determine whether the initiating event will develop into a serious mishap or if the event is sufficiently controlled by the safety systems & procedures implemented in the system design
5. Falls under the system design hazard analysis type (SD-HAT)
6. Worksheet: Event tree diagram (ETD), which provides
 - I. Initiating event
 - II. System pivotal events
 - III. Outcomes
 - IV. Event & outcome probabilities
7. Consequence path: safe operation path, a degraded operation path, & an unsafe operation path
8. Provides a PRA of the risk associated with each potential outcome
9. To model an entire system, with analysis coverage given to subsystems, assemblies, components, software, procedures, environment, & human error
10. I/D?

ETA (Continued)

1. PRA: Risk is based upon a set of triplets: Accident scenarios—what can go wrong?



Accident scenario concept

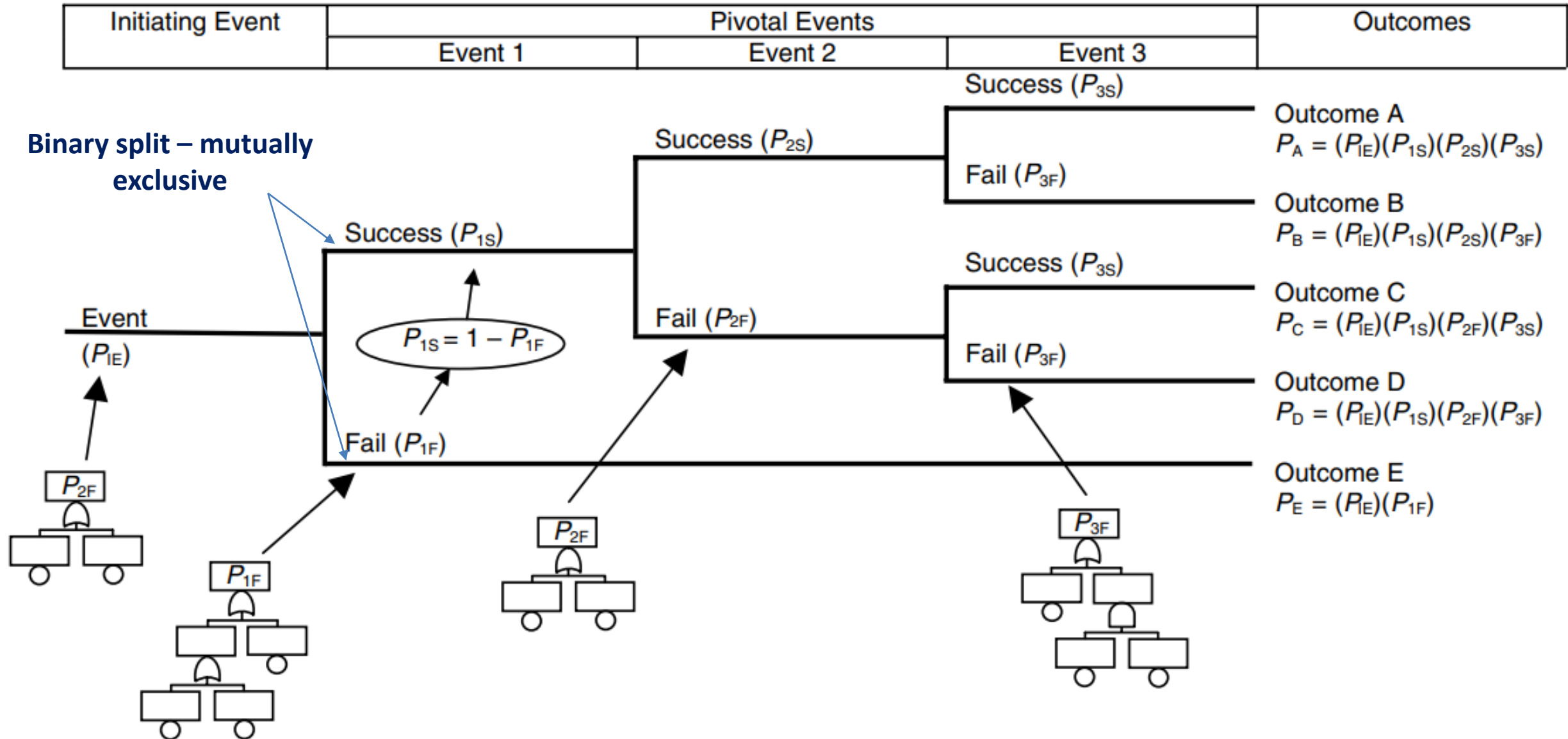


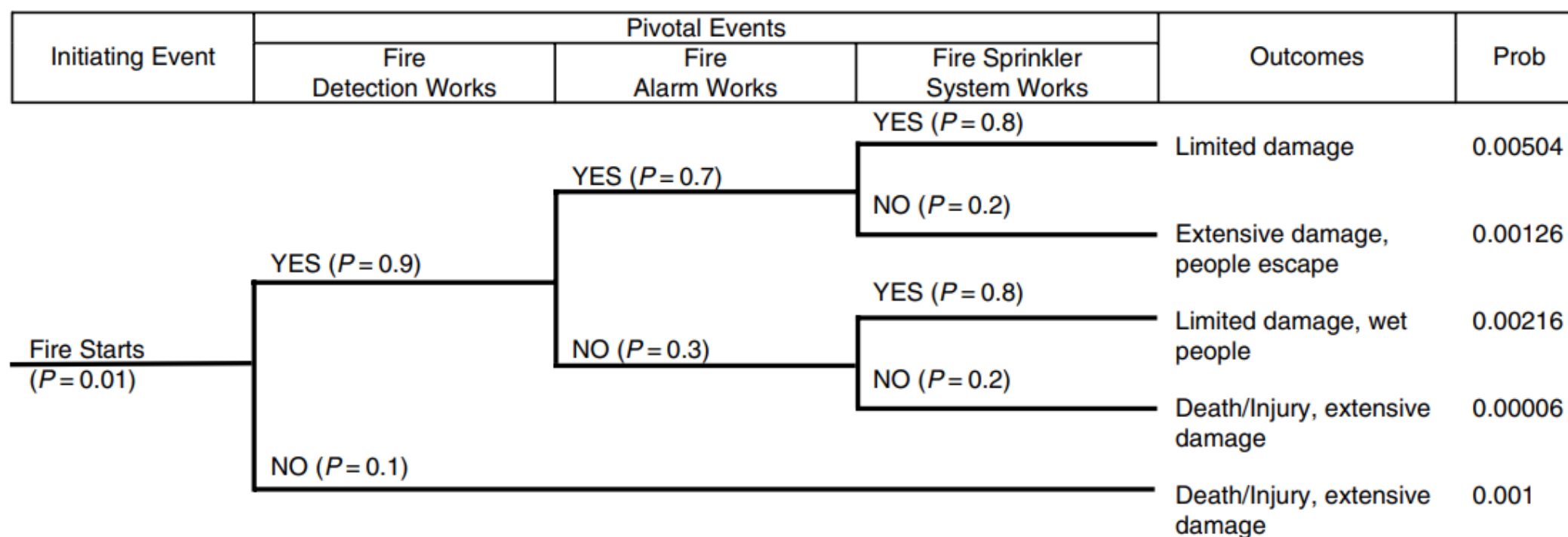
ETA overview

ETA process

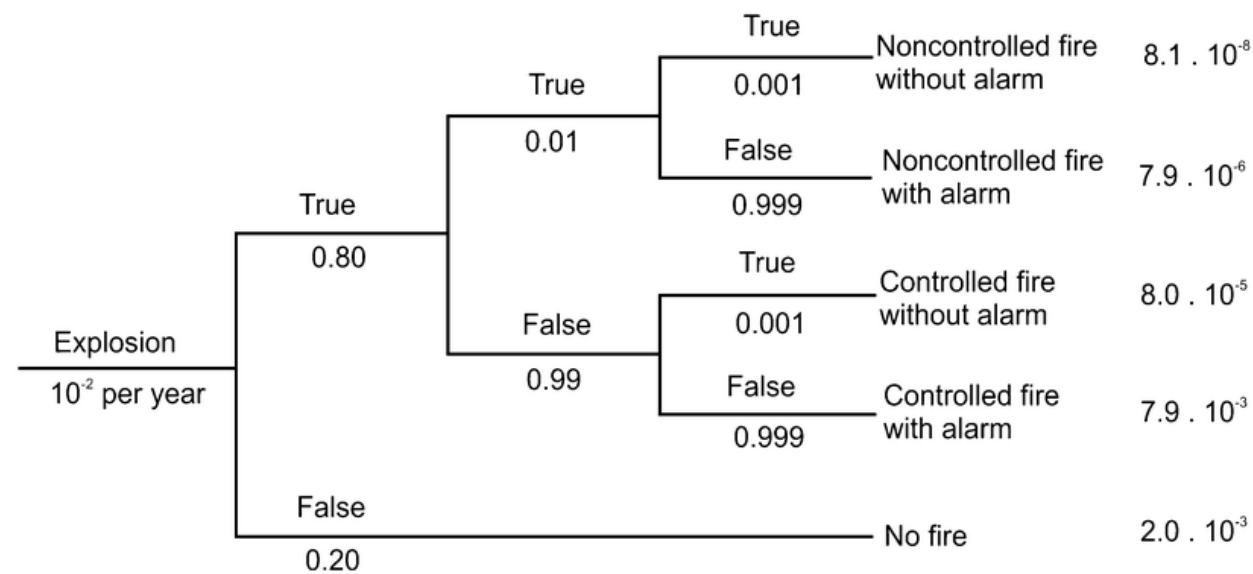
Step	Task	Description
1	Define the system.	Examine the system and define the system boundaries, subsystems, and interfaces.
2	Identify the accident scenarios.	Perform a system assessment or hazard analysis to identify the system hazards and accident scenarios existing within the system design.
3	Identify the initiating events.	Refine the hazard analysis to identify the significant IEs in the accident scenarios. IEs include events such as fire, collision, explosion, pipe break, toxic release, etc.
4	Identify the pivotal events.	Identify the safety barriers or countermeasures involved with the particular scenario that are intended to preclude a mishap.
5	Build the event tree diagram.	Construct the logical ETD, starting with the IE, then the PEs, and completing with the outcomes of each path.
6	Obtain the failure event probabilities.	Obtain or compute the failure probabilities for the PEs on the ETD. It may be necessary to use FTs to determine how a PE can fail and to obtain the probability.
7	Identify the outcome risk.	Compute the outcome risk for each path in the ETD.
8	Evaluate the outcome risk.	Evaluate the outcome risk of each path and determine if the risk is acceptable.
9	Recommend corrective action.	If the outcome risk of a path is not acceptable, develop design strategies to change the risk.
10	Document ETA.	Document the entire ETA process on the ETDs. Update for new information as necessary.

ETD development



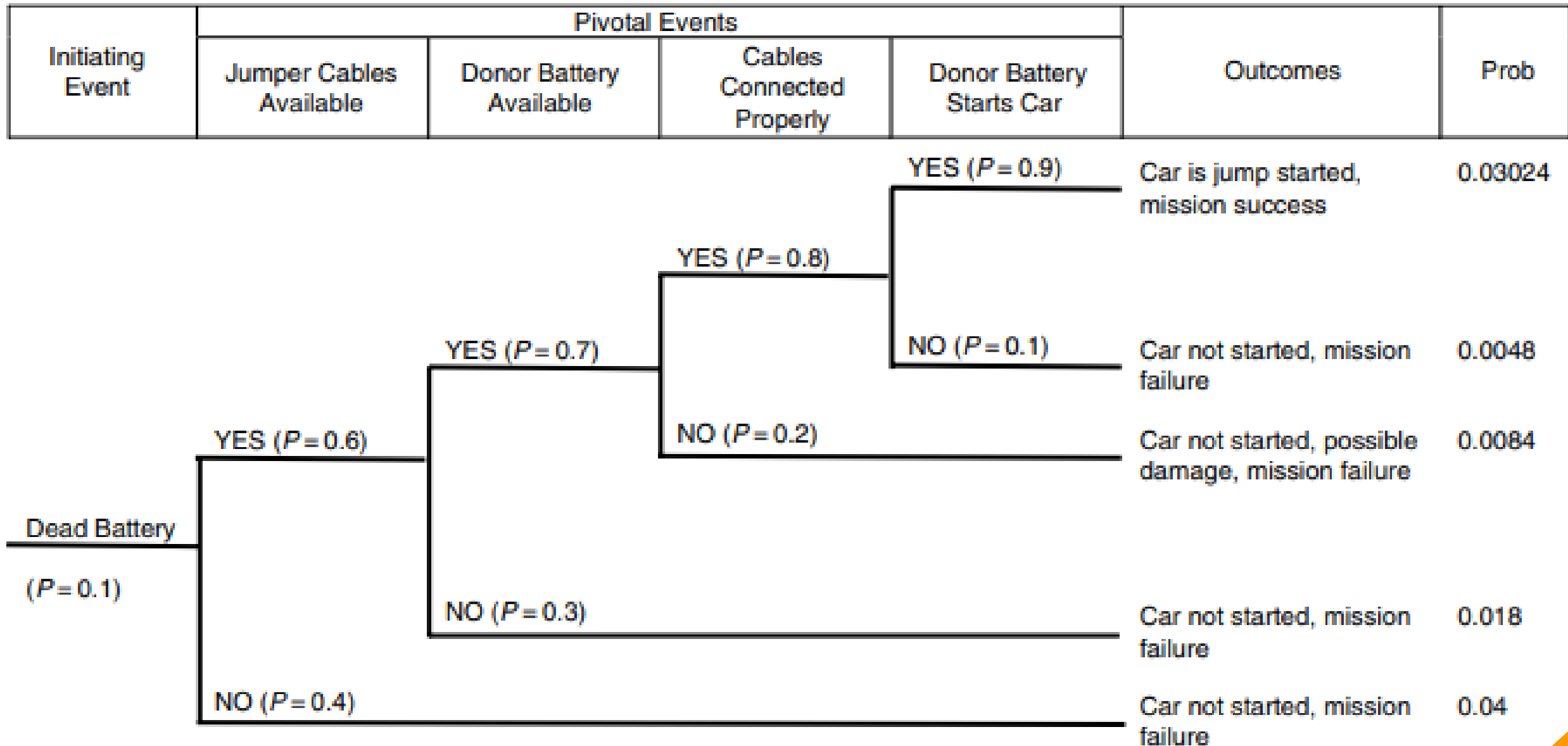


Initiation event	Fire origin	Sprinkler system out of order	Alarm is not activated	Outcomes	Frequency per year
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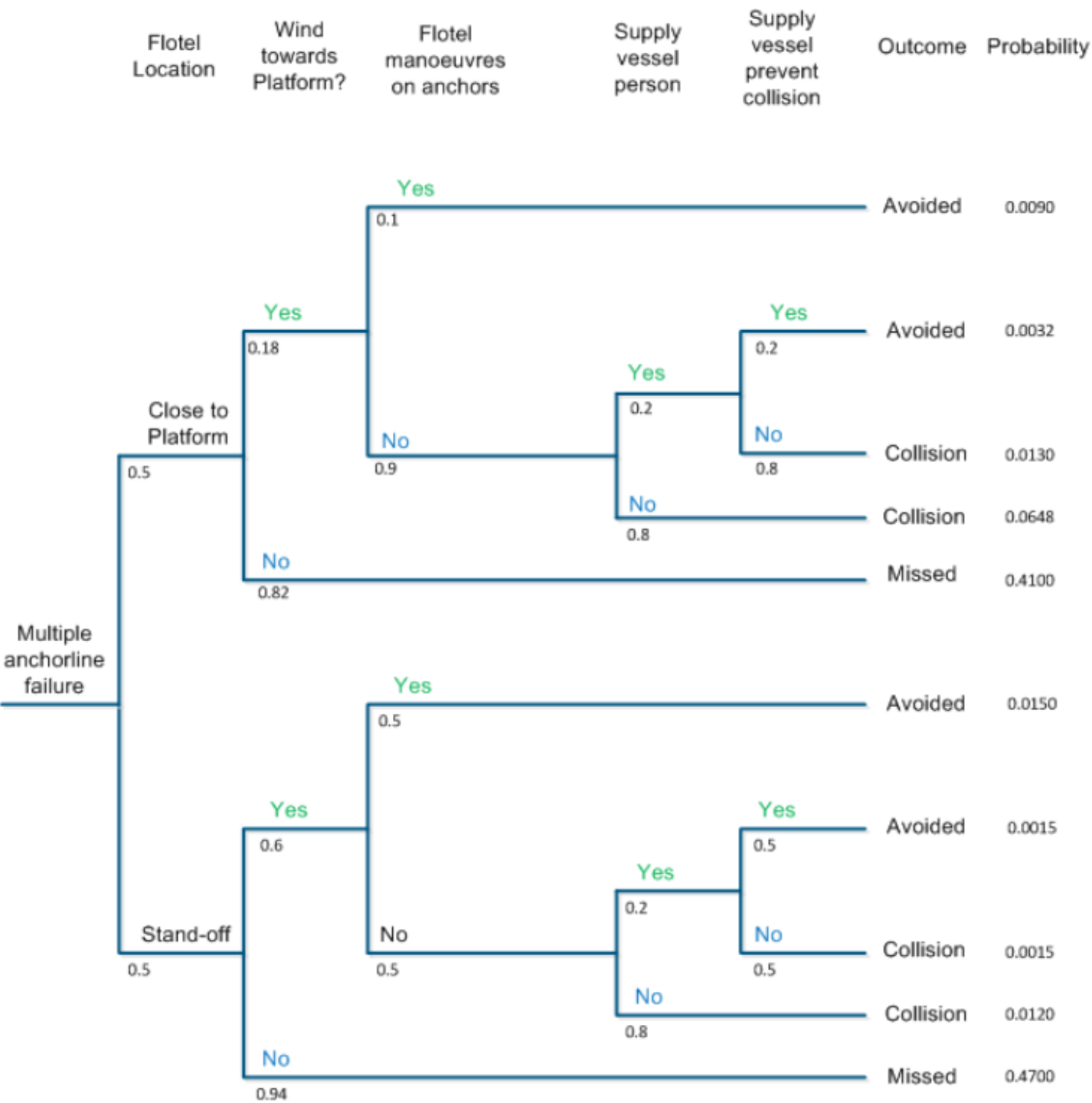
Event tree examples

Event tree example



(<https://www.asems.mod.uk>)

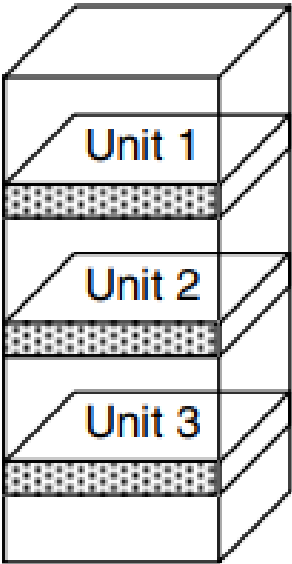
Event tree examples



Failure Mode & Effects Analysis (FMEA)

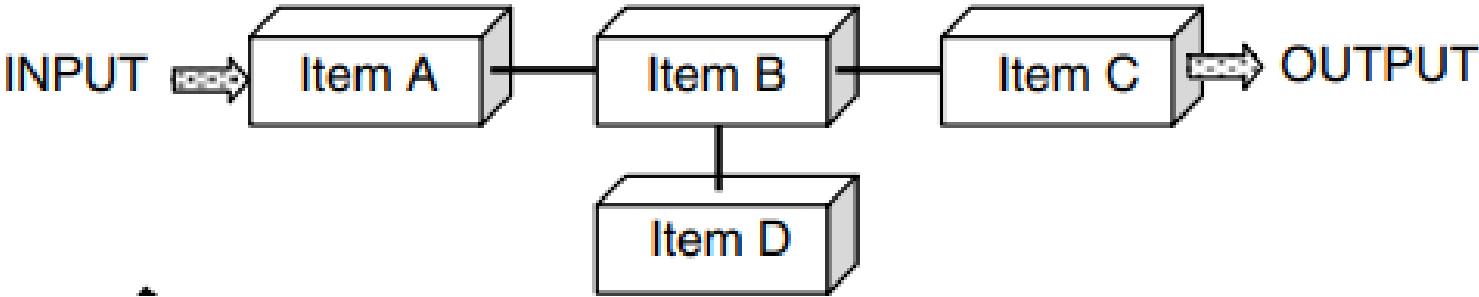
1. Process of identifying potential design weaknesses
2. Reliability improvement tool: for analyzing potential failure modes & calculating subsystem, assembly, or unit failure rates
3. Inductive technique
4. Falls under the (DD-HAT)
5. Manner by which an item fails; the mode or state the item is in after it fails.
6. A relay may fail by
 - a) contacts stuck closed
 - b) contacts slow in opening
 - c) contacts stuck open
 - d) contacts slow in closing
 - e) contact short circuit
 - I. to ground
 - II. to supply
 - III. between contacts
 - IV. to signal lines
 - f) Contacts chattering
 - g) Contacts arcing, generating noise
 - h) Coil open circuit
 - i) Coil short circuit: to supply; to contacts to ground; to signal lines
 - j) Coil resistance: low/high
 - k) Coil overheating
 - l) Coil over magnetized or excessive hysteresis (same effect as contacts stuck closed or slow in opening)
7. Considers every mode of failure of every component
8. Quantitative or qualitative
9. RPN
10. Fault: Undesired anomaly in the functional operation of an equipment or system. The occurrence of an undesired state, which may be the result of a failure
11. FMECA
12. Drawbacks:
 - I. Considers only single item failures & not the combination of items failure
 - II. Does not identify hazards arising from events other than failures

FMEA concept



Subsystem

Unit 1



FMEA Worksheet

Item	Failure Mode	Failure Rate	Immediate Effect	System Effect		
A						
B						
C						
D						

Failure modes

No.	Failure Mode	No.	Failure Mode
1	Structural failure (rupture)	19	Fails to stop
2	Physical binding or jamming	20	Fails to start
3	Vibration	21	Fails to switch
4	Fails to remain (in position)	22	Premature operation
5	Fails to open	23	Delayed operation
6	Fails to close	24	Erroneous input (increased)
7	Fails open	25	Erroneous input (decreased)
8	Fails closed	26	Erroneous output (increased)
9	Internal leakage	27	Erroneous output (decreased)
10	External leakage	28	Loss of input
11	Fails out of tolerance (high)	29	Loss of output
12	Fails out of tolerance (low)	30	Shorted (electrical)
13	Inadvertent operation	31	Open (electrical)
14	Intermittent operation	32	Leakage (electrical)
15	Erratic operation	33	Other unique failure condition
16	Erroneous indication		as applicable to the system
17	Restricted flow		characteristics, requirements
18	False actuation		and operational constraints

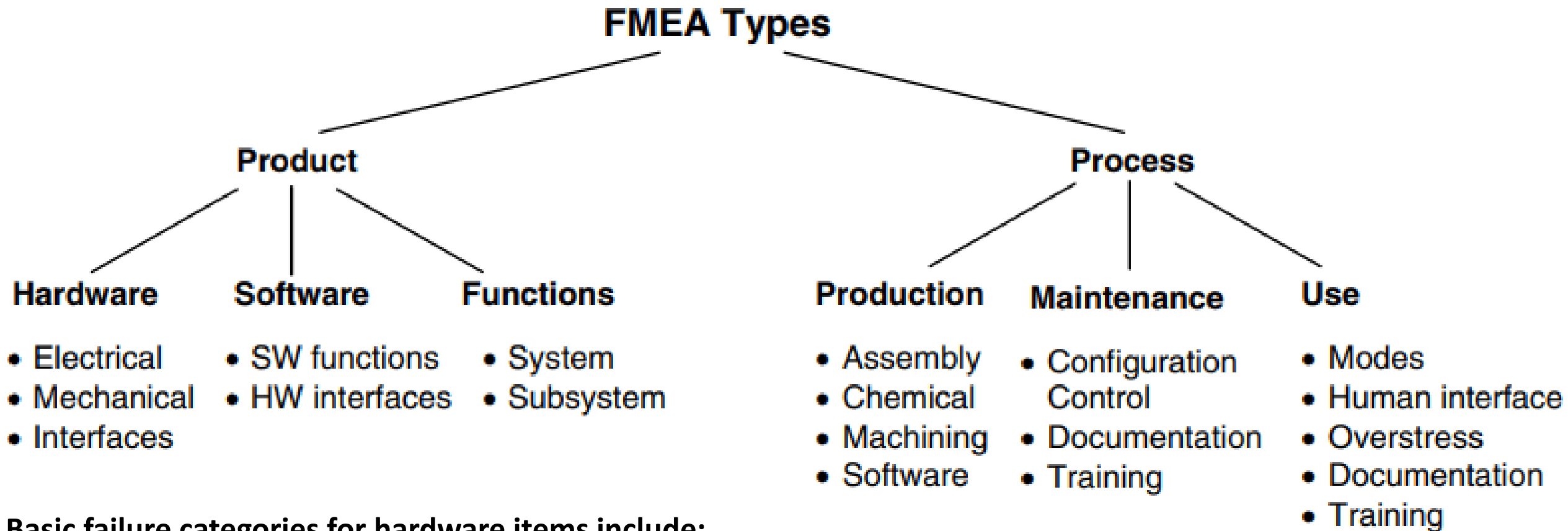
FMEA worksheet - Reliability

Failure Mode and Effects Analysis									
Item	Failure Mode	Failure Rate	Causal Factors	Immediate Effect	System Effect	RPN	Method of Detection	Current Controls	Recomm Action

FMEA worksheet – Safety/Reliability

Failure Mode and Effects Analysis										
System: 1			Subsystem: 2			Mode/Phase: 3				
Item	Failure Mode	Failure Rate	Causal Factors	Immediate Effect	System Effect	Method of Detection	Current Controls	Hazard	Risk	Recomm Action
4	5	6	7	8	9	10	11	12	13	14

FMEA worksheet – Safety/Reliability

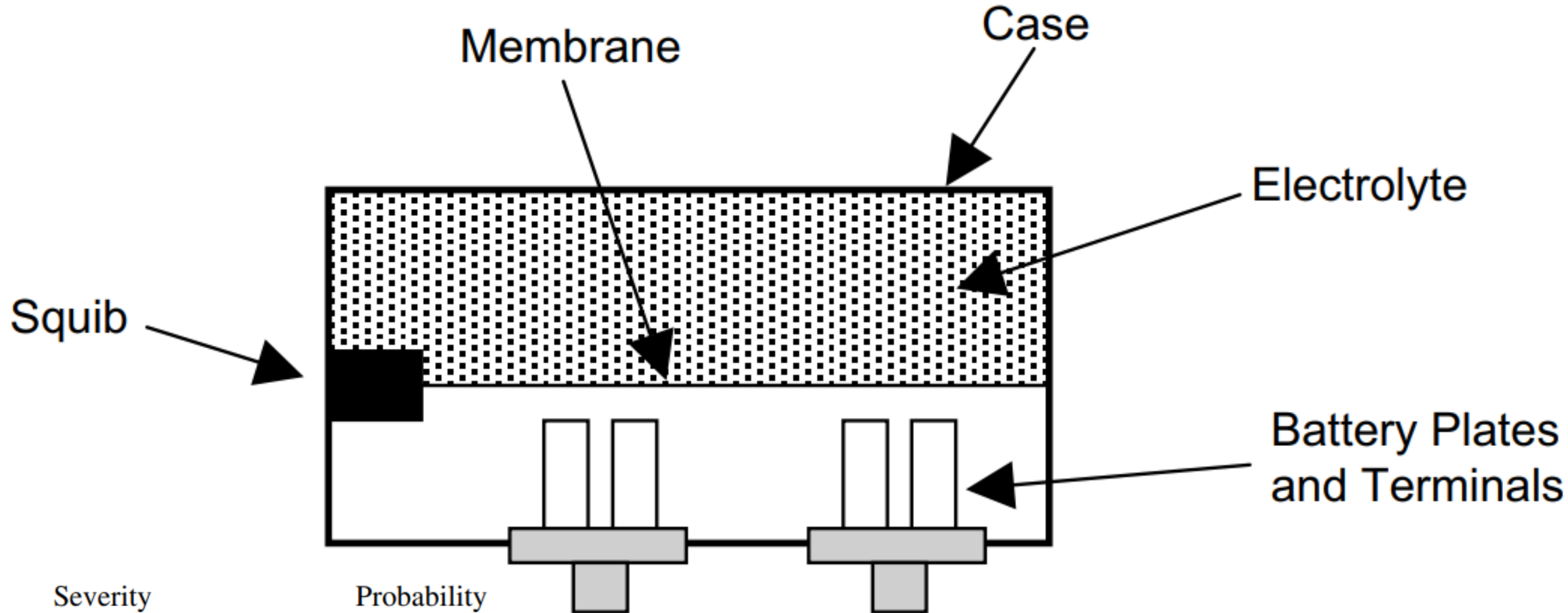


Basic failure categories for hardware items include:

1. Complete failure
2. Partial failure (out of tolerance)
3. Intermittent failure

3 approaches: Functional, Structural, & Hybrid

Example of HFMEA: Missile Battery



Severity	Probability
1. Catastrophic	A. Frequent
2. Critical	B. Probable
3. Marginal	C. Occasional
4. Negligible	D. Remote
	E. Improbable

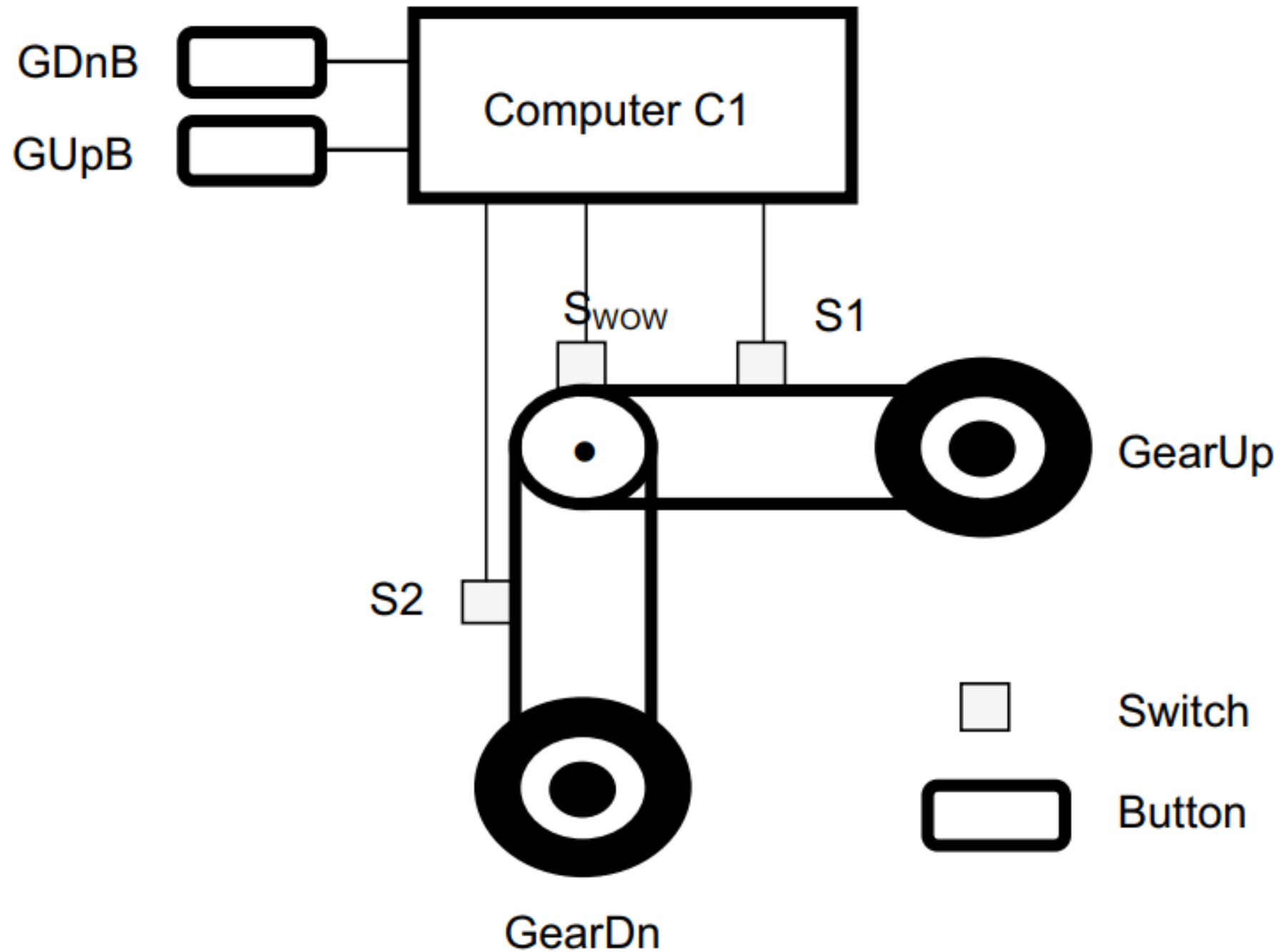
Example: FMEA worksheet

System: Missile			Subsystem: Missile Battery			Mode/Phase: Operation				
Component	Failure Mode	Failure Rate	Causal Factors	Immediate Effect	System Effect	Method of Detection	Current Controls	Hazard	Risk	Recommended Action
Electrolyte	Cracks	3.5×10^{-5} Manuf. data	Manufacturing defect	Electrolyte leakage	No power output from battery	Inspection	QA	Fire source	2D	Add system sensor
	Pinholes	1.1×10^{-9} Manuf. data	Material defect	Electrolyte leakage	No power output from battery	Inspection	QA	Fire source	2E	Add system sensor
	Leaks out of case	4.1×10^{-6} Manuf. data	Case defect; pinholes	Electrolyte leakage	No power output from battery	Inspection	QA	Fire source	2D	Add system sensor
	Wrong electrolyte used	1.0×10^{-5} Manuf. data	Human error	Does not react with battery plates	No power output from battery	Inspection	QA	Unsafe battery reaction	2D	
Battery plates and terminals	Cracks	2.2×10^{-6} Manuf. data	Material defect	Inadequate battery reaction	Insufficient power output from battery	None	None	None	4D	
	Breaks	1.0×10^{-9} Manuf. data	Material defect	Inadequate battery reaction	No power output from battery	None	None	None	4E	

Example: FMEA worksheet (Continued)

System: Missile			Subsystem: Missile Battery			Mode/Phase: Operation				
Component	Failure Mode	Failure Rate	Causal Factors	Immediate Effect	System Effect	Method of Detection	Current Controls	Hazard	Risk	Recommended Action
Membrane	Cracks	3.5×10^{-5} Manuf. data	Materiel defect	Electrolyte leakage	No power output from battery	Inspection	QA	Fire source	2D	Add system sensor
	Pinholes	1.1×10^{-9} Manuf. data	Materiel defect	Electrolyte leakage	No power output from battery	Inspection	QA	Fire source	2E	Add system sensor
	Fails to rupture	4.1×10^{-6} Manuf. data	Materiel defect	No electrolyte to battery plates	No power output from battery	Inspection	QA	None	2D	Add system sensor
Squib	Fails to fire	4.1×10^{-9} Manuf. data	Materiel defect	No electrolyte to battery plates	No power output from battery	Inspection	None	None	2E	
	Fires prematurely	4.1×10^{-9} Manuf. data	Materiel; RF energy; dropped	Premature electrolyte to battery plates	Premature battery activation	System sensor for power	System sensor for power	Premature missile power	2E	Add system sensor

Example: Landing Gear

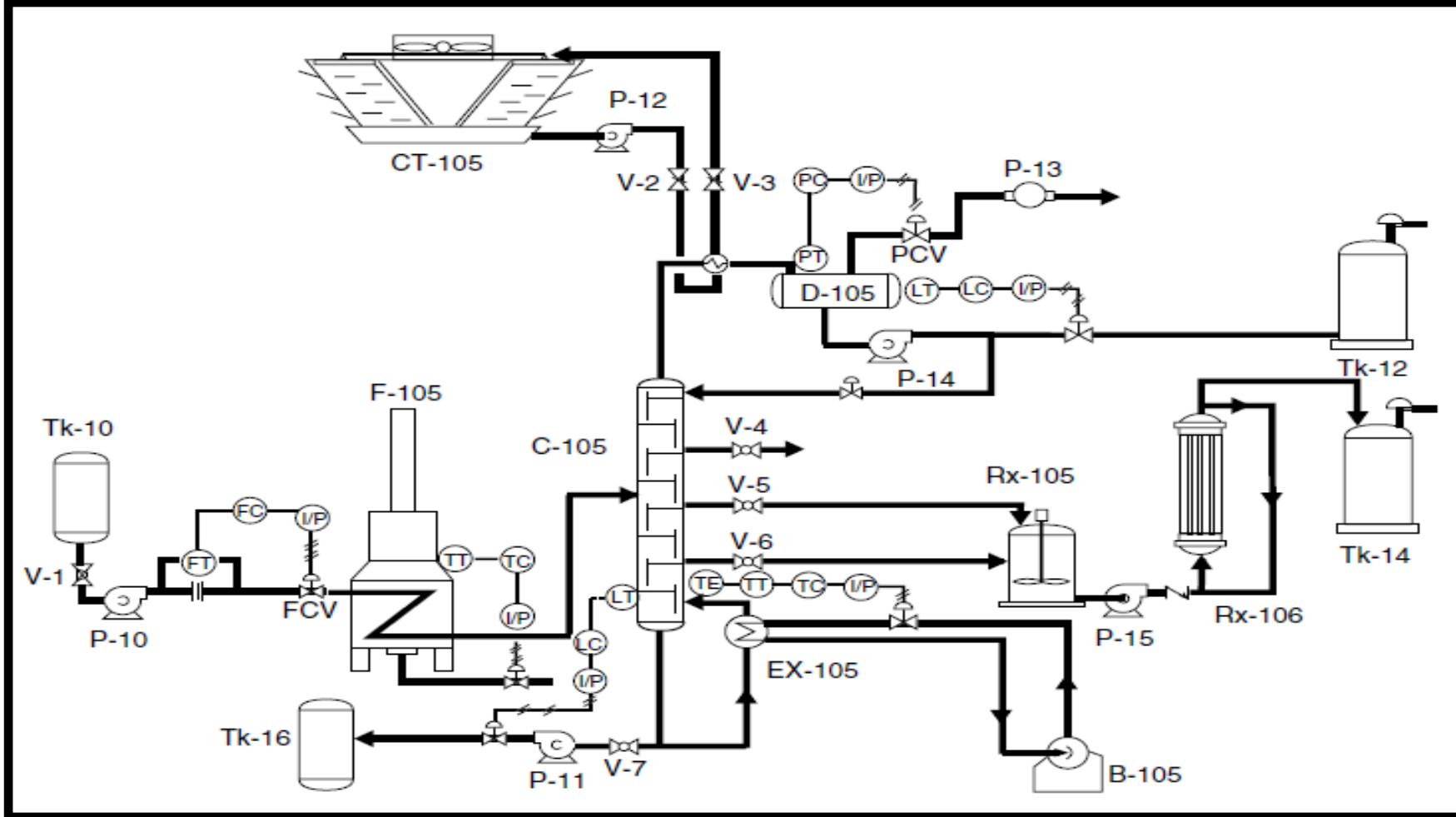


Example: FMEA worksheet

System: Aircraft			Subsystem: Landing Gear					Mode/Phase: Flight		
Function	Failure Mode	Failure Rate	Causal Factors	Immediate Effect	System Effect	Method of Detection	Current Controls	Hazard	Risk	Recommended Action
Gear up	Fails to issue	N/A	Computer; software; wiring	Landing gear	Aircraft flight with gear down	Sensor		Damage from drag	4C	
	Issues prematurely	N/A	Computer; software	If down, landing gear raises prematurely	Gear raises during taxi or takeoff	Sensor		Aircraft damage during taxi	2C	
Gear down	Fails to issue	N/A	Computer; software; wiring	Unable to lower landing gear	Aircraft must land with landing gear up	Sensor		Damage or injury during landing	2C	
	Issues prematurely	N/A	Computer; software	If up, landing gear lowers prematurely	Aircraft flight with gear down	Sensor		Damage from drag	2C	
Gear self-test (BIT = built-in-test)	Fails to occur	N/A	Computer; software; electronic faults	Landing gear fault not detected	Unable to raise or lower gear when desired, no warning	None		Possible unsafe state	4C	
	Occurs erroneously	N/A	Computer; software; electronic faults	Landing gear fault self-test data incorrect	No warning or incorrect warning regarding landing gear status	None		Possible unsafe state	4C	

Example: FMEA worksheet (Continued)

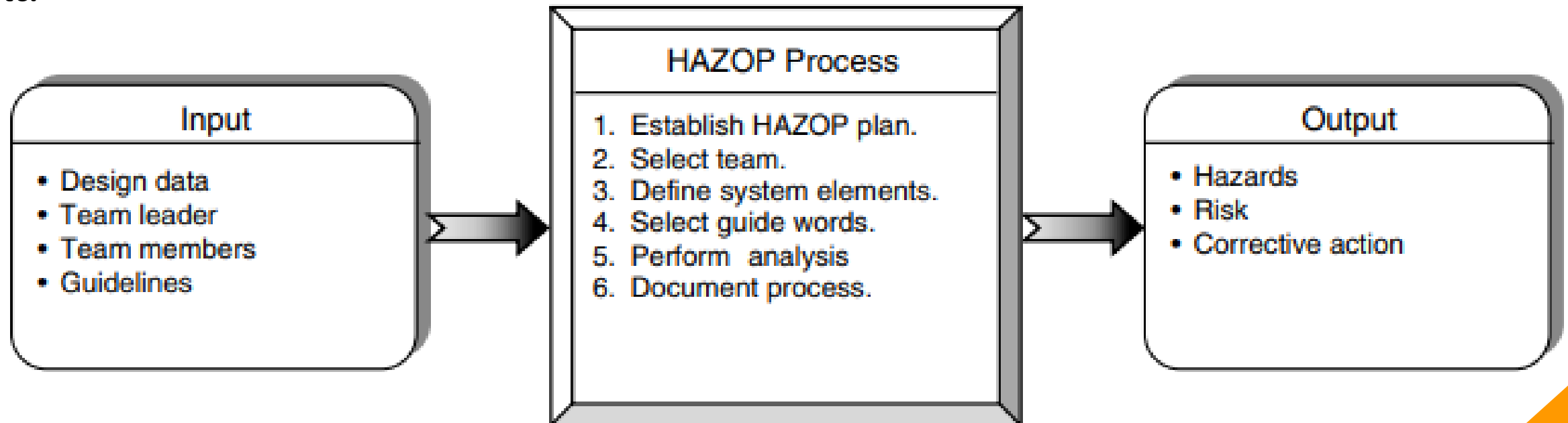
System: Aircraft			Subsystem: Landing Gear				Mode/Phase: Flight			
Function	Failure Mode	Failure Rate	Causal Factors	Immediate Effect	System Effect	Method of Detection	Current Controls	Hazard	Risk	Recommended Action
Report system malfunction	Fails to occur	N/A	Computer; software; electronic faults	BIT data not reported	No warning of fault state	Pilot report		None	4D	
	Occurs erroneously	N/A	Computer; software; electronic faults	BIT data reported incorrectly	No warning of fault state, or incorrect warning	Pilot report		None	4D	
Record self-test results	Fails to occur	N/A	Computer; software; electronic faults	BIT data not recorded	No recording of fault state	Data analysis		None	4D	
	Occurs erroneously	N/A	Computer; software; electronic faults	BIT data recorded incorrectly	No recording of fault state, or incorrect recording	Data analysis		None	4D	



Process Plants

Hazard & Operability (HAZOP)

1. HAZOP analysis = HAZOP study (HAZOPS)
2. Purpose: identify the potential for system deviations from intended operational intent
3. Potential system deviations then lead to possible system hazards
4. Systematically questioning every part of the process to establish how deviations from the design intent can arise - whether such deviations & their consequences can have a negative effect upon the safe & efficient operation of the system
5. Discover credible causes of deviations from design intent – less obvious & not previously observed
6. A team of individuals with expertise in different areas, such as engineering, chemistry, safety, operations, & maintenance
7. Many commercial software packages are available in the market such as SafetyCulture, ProcessMAP, Phast, etc.



Process parameters & guide words for the deviation

Process Parameter	Deviation
Flow	No flow Reverse flow More flow Extra flow Change in flow proportions Flow to wrong place
Temperature	Higher temperature Lower temperature
Pressure	Higher pressure Lower pressure

HAZOPS Vs FMEA

Process parameters & guide words for the deviation

Volume	Higher level (in a tank) Lower level (in a tank) Volume rate changes faster than expected Proportion of volumes is changed
Composition	More component A Less component B Missing component C Composition changed
pH	Higher pH Lower pH Faster change in pH
Viscosity	Higher viscosity Lower viscosity
Phase	Wrong phase Extra phase

Few more process parameters

- Flow (gas, liquid, electric current)
 - Pressure
 - Separate (settle, filter, centrifuge)
 - Reaction
 - Reduce (grind, crush, etc.)
 - Corrode
 - Isolate
 - Vent
 - Inspection, surveillance
 - Viscosity
 - Instruments
 - Corrosion
 - Vibration
 - Software data flow
 - Temperature
 - Level
 - Composition
 - Mix
 - Absorb
 - Erode
 - Drain
 - Purge
 - Maintain
 - Shutdown
 - Startup
 - Erosion
 - Shock
 - Density
-

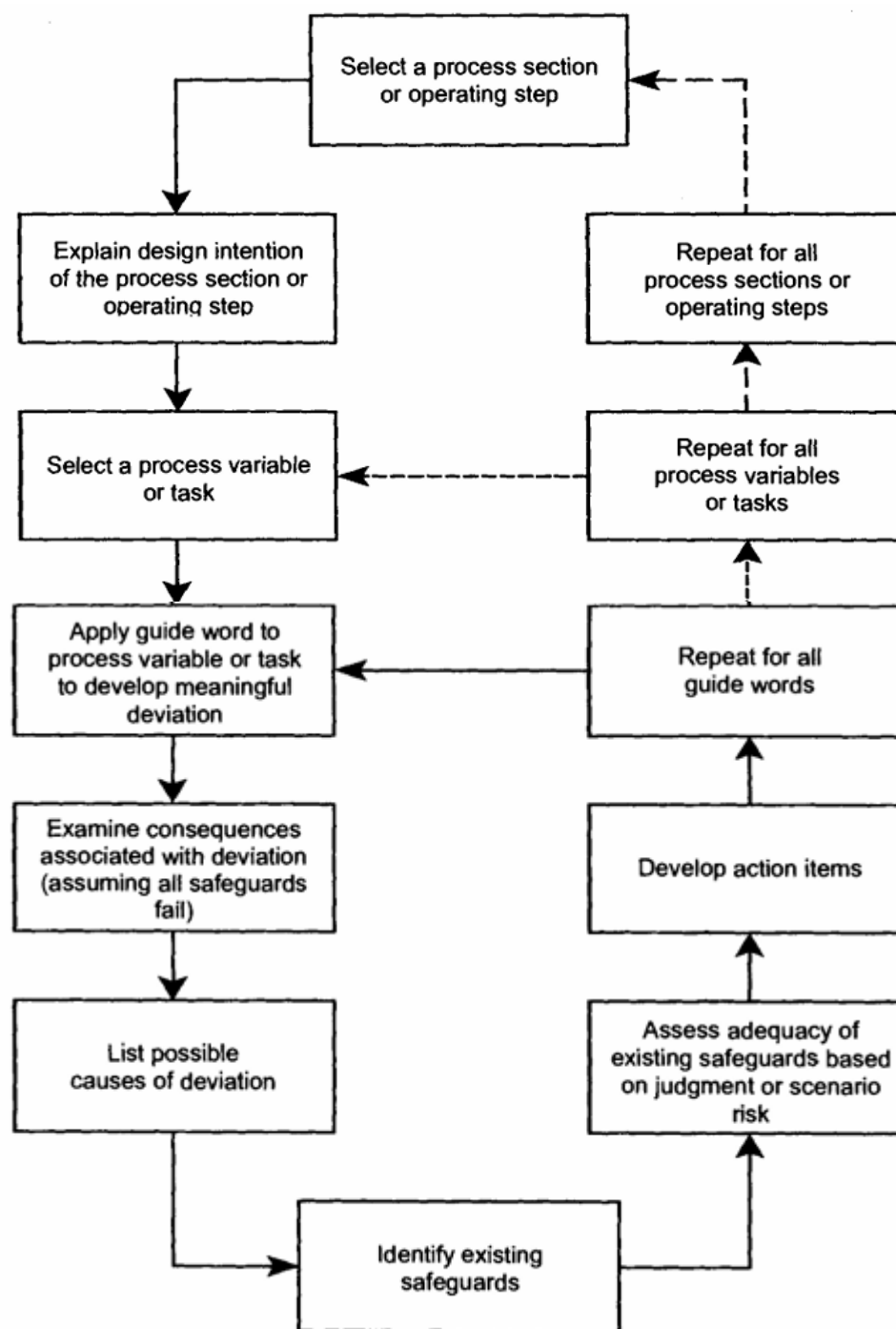
Guide words

Guide Word	Meaning
No	The design intent does not occur (e.g., Flow/No), or the operational aspect is not achievable (Isolate/No).
Less	A quantitative decrease in the design intent occurs (e.g., Pressure/Less).
More	A quantitative increase in the design intent occurs (e.g., Temperature/More).
Reverse	The opposite of the design intent occurs (e.g., Flow/Reverse).
Also	The design intent is completely fulfilled, but in addition some other related activity occurs (e.g., Flow/Also indicating contamination in a product stream, or Level/Also meaning material in a tank or vessel that should not be there).
Other	The activity occurs, but not in the way intended (e.g., Flow/Other could indicate a leak or product flowing where it should not, or Composition/Other might suggest unexpected proportions in a feedstock).
Fluctuation	The design intention is achieved only part of the time (e.g., an air lock in a pipeline might result in Flow/Fluctuation).

Guide words

Guide Word	Meaning
Early	The timing is different from the intention. Usually used when studying sequential operations, this would indicate that a step is started at the wrong time or done out of sequence.
Late	Same as for Early.
As well as (more than)	An additional activity occurs.
Part of	Only some of the design intention is achieved.
Reverse	Logical opposite of the design intention occurs.
Where else	Applicable for flows, transfers, sources, and destinations.
Before/after	The step (or some part of it) is effected out of sequence.
Faster/slower	The step is done/not done with the right timing.
Fails	Fails to operate or perform its intended purpose.
Inadvertent	Function occurs inadvertently or prematurely (i.e., unintentionally).

Flow Diagram



Worksheet

HAZOP Analysis										
No.	Item	Function/ Purpose	Parameter	Guide Word	Consequence	Cause	Hazard	Risk	Recommendation	Comments
1	2	3	4	5	6	7	8	9	10	11

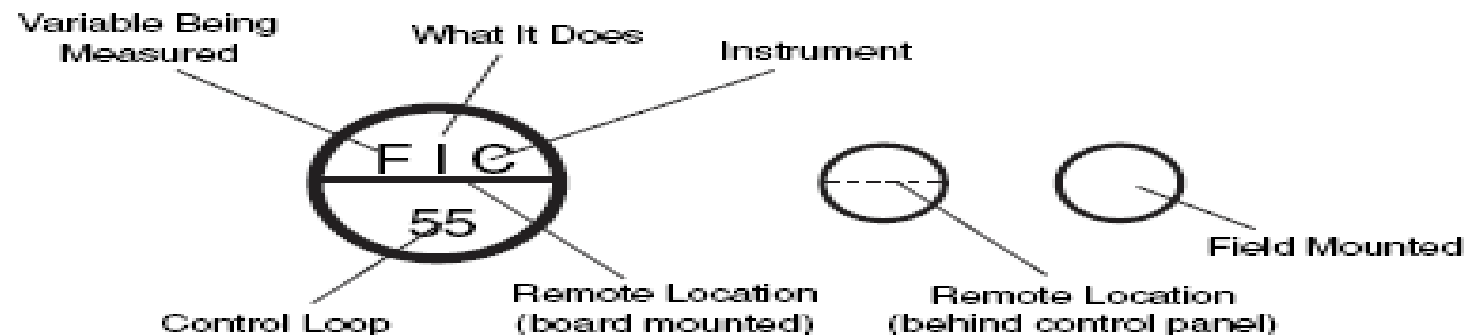
HAZOPS Vs PHA

Please refer Ericson, C. A. (2015). *Hazard analysis techniques for system safety*. John Wiley & Sons for the example of HAZOPS

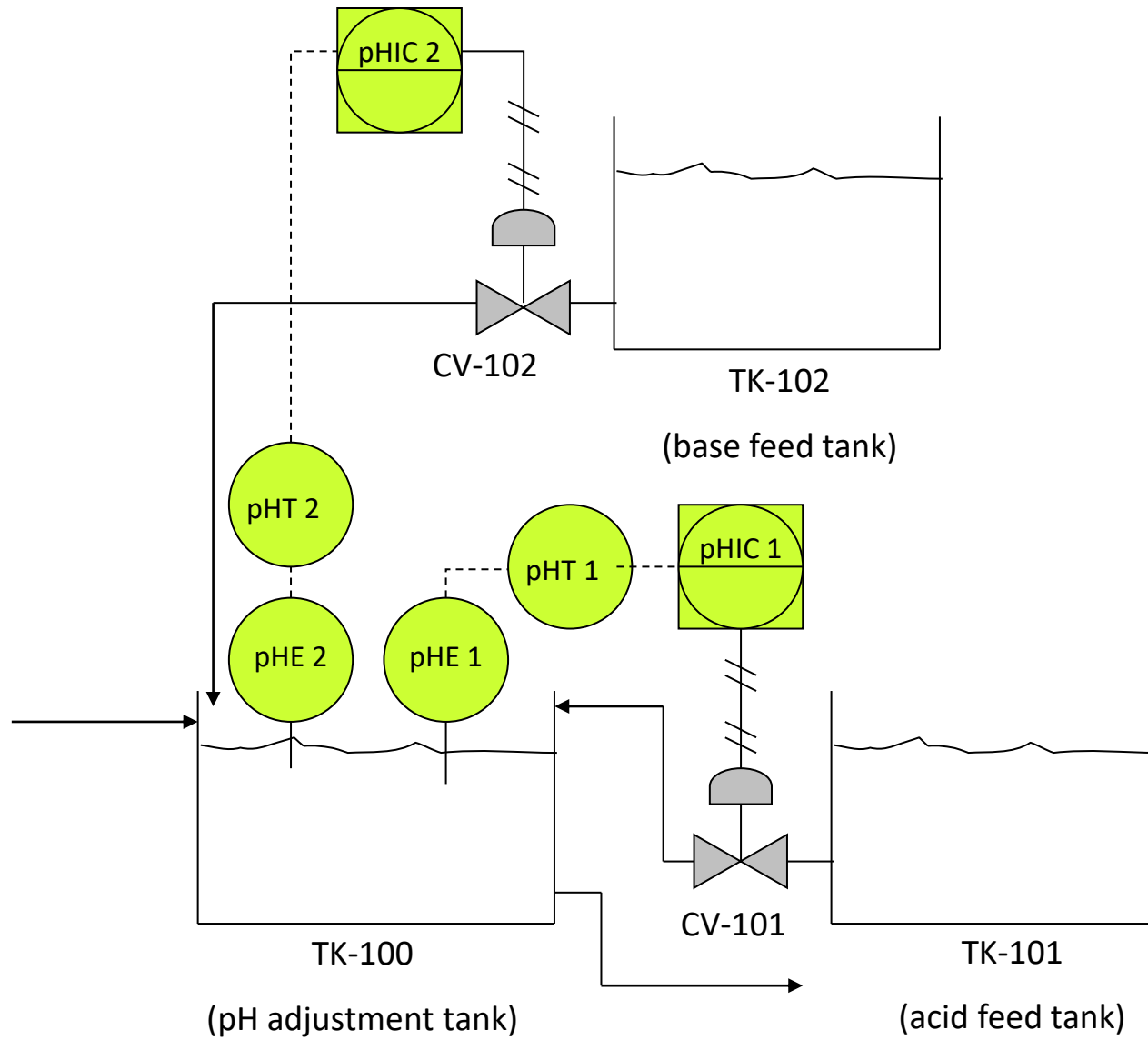
PIPING & INSTRUMENTATION DIAGRAM (P&ID)

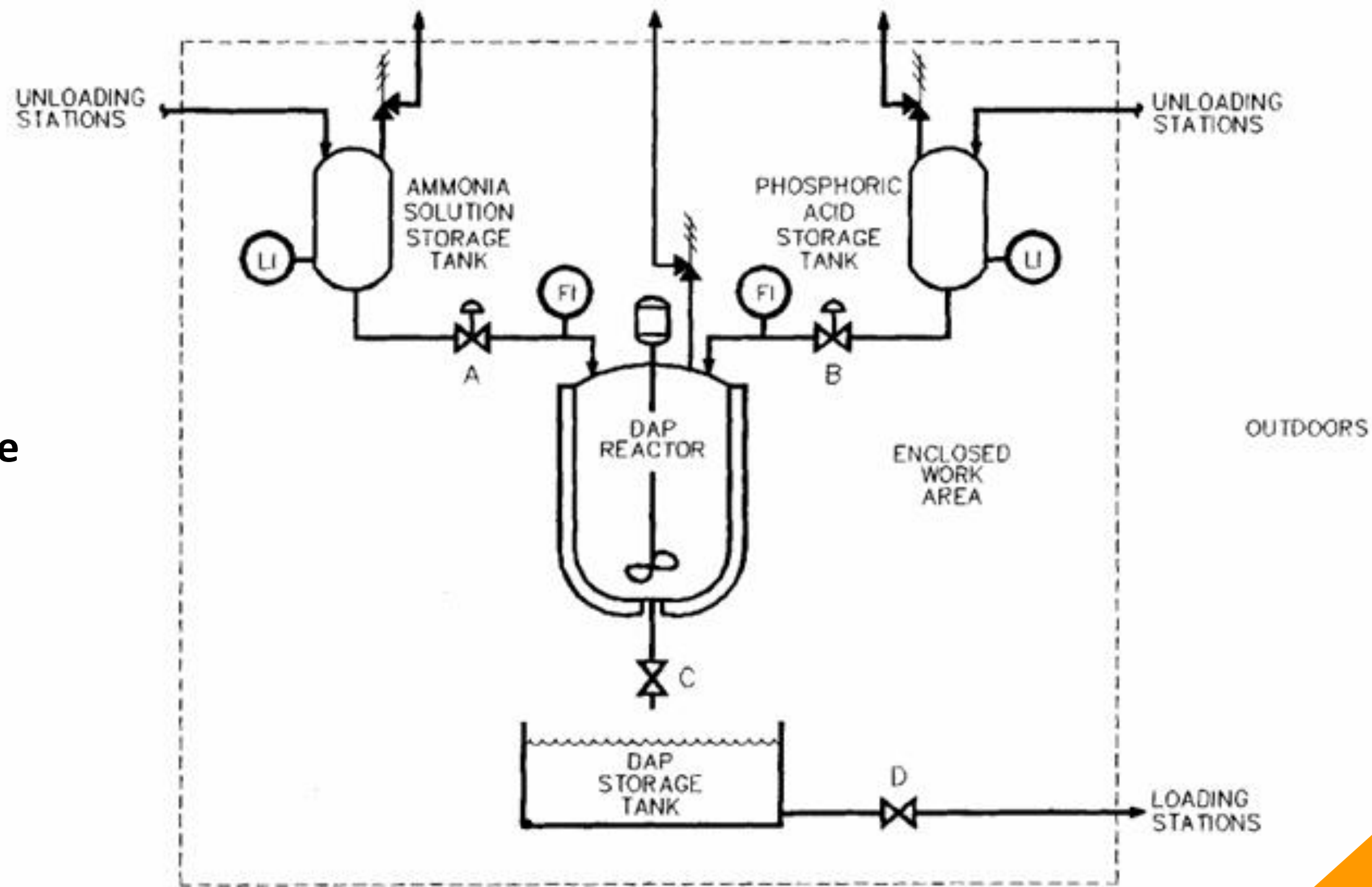
Instrumentation Symbolology

	Temp Indicator		Flow Indicator		Transducer
	Temp Transmitter		Flow Transmitter		Pressure Indicating Controller
	Temp Recorder		Flow Recorder		Pressure Recording Controller
	Temp Controller		Flow Controller		Level Alarm
	Level Indicator		Pressure Indicator		Flow Element
	Level Transmitter		Pressure Transmitter		Temperature Element
	Level Recorder		Pressure Recorder		Level Gauge
	Level Controller		Pressure Controller		Analyzer Transmitter



Example





Example

Example

- A phosphoric acid solution & an ammonia solution are provided through flow control valves to an agitated reactor.
- The ammonia & phosphoric acid react to form diammonium phosphate (DAP), a nonhazardous product.
- The DAP flows from the reactor to an open-top storage tank.
- Relief valves are provided on the storage tanks & the reactor with discharges to outside of the enclosed work area.
- If too much phosphoric acid is fed to the reactor (compared to the ammonia feed rate), an off- specification product is created, but the reaction is safe.
- If the ammonia & phosphoric acid flow rates both increase, the rate of energy release may accelerate, & the reactor, as designed, may be unable to handle the resulting increase in temperature & pressure.
- If too much ammonia is fed to the reactor (as compared to the normal phosphoric acid feed rate), unreacted ammonia may carry over to the DAP storage tank.
- Any residual ammonia in the DAP tank will be released into the enclosed work area, causing personnel exposure.
- Ammonia detectors & alarms are provided in the work area.



Thank you!!



Questions?

